Archive of further tables for "Water-related terms in settlement names, assigned to the nearest grid cells, did not improve within-area predictions of mapped river flood extent: a protocol-based test in 500 Copernicus EMS areas"

Ryota Okauchi and Pang-jo Chun

This file keeps the tables of the exploratory analyses that neither the paper nor its Supplement cites, including tables whose quantities were later corrected or superseded; each caption names the amendment that produced it, and the amendments in the public repository record the corrections. Nothing here is needed to read the paper, and no finding rests on these tables alone. Tables numbered Sk are in the Supplement. Everything outside the frozen protocol is exploratory (Table S5 of the Supplement gives the planning status of each group of estimates).

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Pre-specified analyses

Table A1: Secondary comparisons under each split: the total increment of names over the recalibrated baseline model (including the regional composition of names) and the flexible model refitted with the name features, for the lexicon and learned score and, on the language-model split, for the language-model reading. The 95 % interval is the fixed-margin interval of the decision table (Sect. 2 of the paper); the fourth column gives the interval from resampling the ratio of the two increments, and the fifth the interval of R minus the same comparison with names permuted among the settlements of each AOI.

Split: comparison	R	95 % interval	Ratio bootstrap, 95 %	real − placebo (95 %)	Verdict	TOST
Location groups: total, M1s→M3s	+0.031	[−0.051, +0.155]	[−0.054, +0.192]	[−0.015, +0.030]	inconclusive	no
Location groups: flexible, M1→M3	+0.038	[−0.058, +0.142]	[−0.072, +0.128]	[−0.003, +0.075]	inconclusive	no
Countries: total, M1s→M3s	−0.037	[−0.149, +0.043]	[−0.275, +0.089]	[−0.052, +0.074]	inconclusive	no
Countries: flexible, M1→M3	+0.040	[−0.047, +0.158]	[−0.079, +0.391]	[−0.143, +0.077]	inconclusive	no
Time: total, M1s→M3s	+0.007	[−0.101, +0.113]	[−0.329, +0.328]	[−0.149, +0.111]	undefined-denominator	—
Time: flexible, M1→M3	−0.128	[−0.513, +0.215]	[−2.026, +1.544]	[−0.168, +0.636]	undefined-denominator	—
Language-model cut-off: total, M1s→M3s	+0.001	[−0.025, +0.028]	[−0.026, +0.087]	[−0.065, +0.022]	equivalent-null	yes
Language-model cut-off: flexible, M1→M3	+0.168	[+0.039, +0.328]	[+0.040, +1.126]	[−0.0004, +0.068]	not-meaning	no
Language-model cut-off: language model, total, M1s→M3rs	+0.027	[+0.0003, +0.056]	[−0.001, +0.195]	[+0.007, +0.045]	positive-small	no
Language-model cut-off: language model, flexible, M1→M3r	+0.052	[−0.023, +0.138]	[−0.034, +0.380]	[−0.161, +0.014]	inconclusive	no

Table A2: Ratio-bootstrap 95 % intervals of R for every comparison and split (post hoc, planned in amendment 2): the two increments are resampled together (cluster bootstrap over location groups, countries for the country hold-out, 4,000 replicates) and their ratio taken in each replicate. The decision table uses fixed-margin intervals instead (Sect. 2 of the paper), because the ratio mixes the uncertainty of the denominator into R. [u]The pre-specified rule leaves R undefined on this split. —: comparison not run on this split.

Comparison	Location groups	Countries	Time[u]	LM cut-off	Events grouped
Names within areas, M1c→M3n (H2)	[−0.016, +0.004]	[−0.008, +0.025]	[−0.089, +0.069]	[−0.014, +0.004]	[+0.001, +0.029]
Names permuted, M1c→M3np	[−0.035, +0.003]	[−0.046, +0.006]	[−0.297, +0.186]	[−0.034, +0.028]	[−0.015, +0.088]
Lexicon only, M1c→M3nx	[−0.019, +0.002]	[−0.002, +0.038]	[−0.113, +0.063]	[−0.015, +0.004]	—
Learned score only, M1c→M3nl	[−0.006, +0.001]	[−0.004, +0.028]	[−0.029, +0.039]	[−0.008, +0.002]	—
Position only, M0c→M0n (H1)	[−0.006, +0.002]	[−0.006, +0.020]	[−0.025, +0.025]	[−0.006, +0.008]	—
Position only, flexible, M0→M0N	[+0.006, +0.252]	[+0.042, +0.287]	[−7.715, +5.075]	[−0.187, +0.011]	—
Total, M1s→M3s	[−0.054, +0.192]	[−0.275, +0.089]	[−0.329, +0.328]	[−0.026, +0.087]	[−0.144, +0.093]
Flexible, M1→M3	[−0.072, +0.128]	[−0.079, +0.391]	[−2.026, +1.544]	[+0.040, +1.126]	—
Language model, M1rc→M3rn (H4)	—	—	—	[−0.034, +0.013]	—
Language model, permuted	—	—	—	[−0.028, +0.003]	—
Language model, total, M1s→M3rs	—	—	—	[−0.001, +0.195]	—
Language model, flexible	—	—	—	[−0.034, +0.380]	—
Names centred within AOIs, M1s→M3d	[−0.012, −0.0004]	[−0.014, +0.0001]	[−0.041, +0.028]	[−0.005, +0.009]	[−0.005, −0.001]

Table A3: Operating characteristics of the decision rule on synthetic labels for each split (the primary split before the freeze; the others post hoc, amendment 2), with 95 % Wilson intervals for the Monte-Carlo error. The simulated effect is δ times the indicator that the nearest name is water-related in the lexicon, calibrated so that $R \approx 0.05$ on the primary split ($\delta = 1.2$, an odds ratio of 3.3); on the other splits the same δ gives the mean R shown, not the smallest effect of interest. Equivalent counts the replicates whose 90 % interval of R lies within ± 0.05, whatever their label. The last column gives, for $\delta = 1.2$, the share of replicates whose 90 % interval of ΔBS lies within the equivalence margin of the real labels on the same split (post hoc, amendment 2): how often an effect of this size would have been called equivalent in our data (—: margin undefined on the time split). Margin ratio and upper ratio (amendment 2, addendum 3): the real-label equivalence margin and the upper end of the observed 95 % interval of ΔBS on the same split, each divided by the mean ΔBS that $\delta = 1.2$ produced there; these ratios ignore that intervals widen under the alternative, so the column before them is the better guide. [n]Includes replicates whose simulated baseline gain was negative (dividing by a negative increment reverses the interval of R). [a]The effect is injected through the lexicon, so this row is the lexicon test on the language-model split; there is no power analysis for the language-model reading (H4).

Split	δ	Replicates	Equivalent (%)	Positive (%)	Mean R	Labels	Equivalent, real margin (%)	Margin ratio	Upper ratio
Location groups (pre-specified)	0	100	99 [95, 100]	0 [0, 4]	−0.001	equivalent-null 91, inconclusive 1, negative 7, not-meaning 1	—	—	—
	1.2	100	0 [0, 4]	98 [93, 99]	+0.057	inconclusive 1, positive-meaningful 69, positive-small 29, undefined-denominator 1	26 [18, 35]	1.5	0.11
Countries	0	20	95 [76, 99]	0 [0, 16]	−0.004	equivalent-null 16, inconclusive 1, negative 3	—	—	—
	1.2	20	0 [0, 16]	65 [43, 82]	+0.081	not-meaning 4, positive-meaningful 12, positive-small 1, undefined-denominator 3	0 [0, 16]	1.1	0.25
Time	0	20	100 [84, 100]	0 [0, 16]	+0.004	equivalent-null 12, negative 1, undefined-denominator 7	—	—	—
	1.2	20	65 [43, 82]	5 [1, 24]	+0.017	equivalent-null 10, inconclusive 2, positive-negligible 1, undefined-denominator 7	—	—	0.99
Language-model cut-off, lexicon effect[a]	0	20	100 [84, 100]	0 [0, 16]	+0.0002	equivalent-null 18, negative 1, undefined-denominator 1	—	—	—
	1.2	20	85 [64, 95]	15 [5, 36]	+0.013	equivalent-null 13, inconclusive 2, positive-negligible 3, undefined-denominator 2	100 [84, 100][n]	4.2	0.18

Table A4: Secondary metrics on the primary split (pooled over held-out folds).

Model	Brier	Log loss	PR-AUC	Top-10 % capture (%)
M1 (baseline model)	0.04374	0.15218	0.327	51.0
M3 (refitted with names)	0.04346	0.15161	0.332	51.3
M1s (recalibrated M1)	0.04337	0.15159	0.325	50.9
M3s (+ names)	0.04314	0.15105	0.334	51.7
M1c (+ regional composition)	0.04309	0.15104	0.348	52.9
M3n (+ names within areas)	0.04312	0.15113	0.348	52.9

Table A5: The confirmatory analysis with the frozen code (lexicon not applied to Norwegian names) and with the corrected reading (amendment 4): R of the within-area comparison, 95 % interval and verdict.

Split	R (frozen)	95 %	Verdict	R (corrected)	95 %	Verdict
location groups held out (primary)	−0.003	[−0.013, +0.004]	equivalent-null	−0.004	[−0.015, +0.003]	equivalent-null
countries held out	+0.001	[−0.005, +0.011]	equivalent-null	+0.0005	[−0.003, +0.005]	equivalent-null
time held out	−0.001	[−0.027, +0.023]	undefined-denominator	−0.013	[−0.039, +0.008]	undefined-denominator
after the language model's cut-off	−0.002	[−0.007, +0.002]	equivalent-null	−0.003	[−0.011, +0.003]	equivalent-null

Table A6: Second-stage coefficients on the standardised scale (post hoc, amendment 5): mean over the ten folds of the primary split, with the smallest and largest fold in parentheses. M3n: the name features; M3np: the same features of names reassigned among settlements; M3d: the features centred within each AOI; M1c: the regional composition (AOI means). The reading of Norwegian names is the corrected one (amendment 4). The indicator that the lexicon covers a language of the country is one for every analysed cell with the real names, so its standardised coefficient is zero and the second stage has effectively fourteen name features.

Feature	M3n	M3np (placebo)	M3d	M1c (composition)
logit of the terrain prediction	+1.925 (+1.769, +2.088)	+1.924 (+1.778, +2.085)	+1.969 (+1.883, +2.092)	+1.924 (+1.769, +2.087)
water, nearest name	+0.006 (−0.009, +0.015)	+0.051 (−0.001, +0.108)	+0.003 (−0.003, +0.011)	−0.015 (−0.242, +0.076)
wetland, nearest name	+0.001 (−0.007, +0.006)	−0.023 (−0.030, −0.017)	+0.001 (−0.004, +0.005)	−0.058 (−0.164, +0.083)
riverine, nearest name	−0.003 (−0.012, +0.001)	+0.027 (−0.004, +0.040)	−0.002 (−0.010, +0.005)	+0.145 (+0.003, +0.231)
crossing, nearest name	+0.001 (−0.013, +0.009)	−0.018 (−0.031, −0.004)	+0.005 (−0.007, +0.010)	+0.014 (−0.130, +0.156)
lowland, nearest name	−0.004 (−0.022, +0.011)	−0.025 (−0.042, −0.010)	−0.003 (−0.013, +0.008)	−0.260 (−0.365, −0.170)
highland, nearest name	+0.010 (−0.012, +0.027)	−0.033 (−0.051, −0.005)	+0.010 (−0.004, +0.027)	+0.131 (+0.017, +0.382)
water, three nearest (weighted)	−0.036 (−0.061, −0.019)	−0.059 (−0.089, −0.038)	−0.014 (−0.027, −0.000)	−0.271 (−0.453, +0.098)
wetland, three nearest (weighted)	−0.004 (−0.011, +0.004)	−0.020 (−0.043, −0.006)	−0.007 (−0.013, +0.001)	−0.073 (−0.234, +0.083)
riverine, three nearest (weighted)	+0.023 (+0.003, +0.035)	−0.017 (−0.028, −0.004)	+0.025 (+0.001, +0.038)	−0.037 (−0.174, +0.222)
crossing, three nearest (weighted)	−0.029 (−0.041, −0.016)	+0.003 (−0.008, +0.012)	−0.028 (−0.040, −0.015)	−0.033 (−0.138, +0.162)
lowland, three nearest (weighted)	−0.045 (−0.060, −0.025)	−0.045 (−0.082, −0.015)	−0.032 (−0.041, −0.023)	+0.213 (+0.090, +0.324)
highland, three nearest (weighted)	+0.019 (−0.012, +0.031)	−0.047 (−0.073, −0.012)	+0.023 (−0.004, +0.038)	+0.136 (−0.143, +0.234)
water-related, nearest name	+0.013 (+0.006, +0.023)	+0.037 (+0.015, +0.057)	+0.007 (+0.002, +0.013)	−0.123 (−0.195, −0.067)
lexicon covers the language	+0.000 (+0.000, +0.000)	+0.007 (+0.000, +0.011)	+0.000 (+0.000, +0.000)	+0.000 (+0.000, +0.000)
character-model score	+0.010 (−0.008, +0.026)	−0.017 (−0.090, +0.020)	+0.007 (−0.006, +0.019)	+0.224 (+0.175, +0.292)

Table A7: Interval coverage in the pre-specified null simulations (amendment 2), the p values required by the protocol (amendment 6, addendum 1), the spatial correlation of held-out residuals and reference points for the baseline model (primary split). The CSI maps use the observed flooded count of each AOI and are therefore optimistic. TOST p: no bootstrap replicate fell outside the margin, so p is below $1/B$ for B replicates (the value in brackets).

Quantity	Value
Null simulations: coverage of the 90 % / 95 % interval (mean R)	92 % / 97 %
Null simulations: coverage of zero, 90 % / 95 %	84 % / 92 %
Null simulations: between-replicate SD of R (robust SD; outlying replicates); median bootstrap SE of R	0.0045 (0.0021; 13); 0.0021
Null simulations: verdicts	equivalent-null 91, inconclusive 1, negative 7, not-meaning 1
H2: TOST p	$< 1/B$ (0.00025)
H3: one-sided p ($\Delta_0 > \Delta_4$); Spearman ρ and exact one-sided p	0.86; -0.2, 0.39
H4: TOST p; one-sided p for an improvement	$< 1/B$ (0.00025); 0.66
Holm-adjusted p, H3 / H4	1.00 / 1.00
Correlation of held-out M1 residuals within AOIs by distance (km)	0-0.5: 0.50; 0.5-1: 0.29; 1-2: 0.15; 2-5: 0.04; 5-10: 0.00; 10-20: -0.02
CSI pooled over all cells of maps flagging the observed flooded share per AOI: mean HAND (Table S18) / M1 / M3n	0.25 / 0.30 / 0.30
Brier score: constant (training rate) / M0 / M1	0.0523 / 0.0513 / 0.0437

Exploratory checks of the verdict

Table A8: Corrected equivalence test on the ratio (exploratory, amendment 13, item B1). $L_{\pm} = \Delta \mp 0.05D$ with Δ the change in Brier score from names and D the baseline gain, computed on the same resamples; equivalence when the one-sided 95 % upper limit of L_+ lies below zero and the lower limit of L_- above zero. Fieller: the 90 % interval of R from the variances and covariance of the resamples (normal approximation). (a) 4,000 resamples of location groups with the fitted models held fixed; (b) 500 resamples of location groups with the second stage refitted (corrected code); (c) 500 two-stage resamples, countries then activations within each drawn country, with the second stage refitted; (d) (c) plus the variance of $L_{\pm}$ over 13 assignments of location groups to folds, corrected for overlapping training sets (Nadeau–Bengio; normal approximation). The lower block repeats the double resampling of the protocol rule's operating characteristics for the corrected test (a).

Resampling	R	L_+ upper	L_- lower	Fieller 90 %	Equivalent
(a) Confirmatory run, models fixed	-0.003	-2.4×10^{-4}	1.8×10^{-4}	-0.013 to 0.003	yes
(a) Corrected code, models fixed	-0.004	-2.5×10^{-4}	1.7×10^{-4}	-0.015 to 0.004	yes
(a) H4, nested	-0.003	-1.7×10^{-4}	9.8×10^{-5}	-0.028 to 0.010	yes
(a) H4, total	0.027	2.5×10^{-4}	3.8×10^{-4}	0.003 to 0.161	no
(a) Countries held out	0.001	-6.5×10^{-5}	9.2×10^{-5}	-0.008 to 0.022	yes
(b) Location groups, second stage refitted	-0.004	-2.1×10^{-4}	1.7×10^{-4}	-0.025 to 0.016	yes
(c) Activations within countries, refitted	-0.004	-1.9×10^{-4}	1.1×10^{-4}	-0.042 to 0.031	yes
(d) As (c), with fold-assignment variance	—	-9.6×10^{-5}	3.9×10^{-5}	-0.042 to 0.031	yes
Share of samples declared equivalent with the true R set to -0.05 / 0 / $+0.05$ (models fixed; Sect. 3.2 of the paper)					
protocol's rule, uniform shift	32 % / 99.9 % / 31 %				
corrected test (a), uniform shift	3.1 % / 98.6 % / 2.6 %				
protocol's rule, shift proportional to the baseline gain	0.0 % / 99.9 % / 0.0 %				
corrected test (a), shift proportional to the baseline gain	15 % / 98.6 % / 2.7 %				
Other measures: R over a constant forecast (Y3), the change in Brier units and the increment in Brier skill					
Primary, confirmatory run	-0.003	ΔBS -2.6×10^{-5}		skill -5.1×10^{-4}	
Primary, corrected code	-0.004	ΔBS -3.1×10^{-5}		skill -5.8×10^{-4}	
H4, nested	-0.015	ΔBS -3.8×10^{-5}		skill -7.3×10^{-4}	
H4, total	0.116	ΔBS 3.0×10^{-4}		skill 5.7×10^{-3}	

Table A9: Models of the post hoc analyses. All use the second stage of the primary comparison (Table 1 of the paper); the amendment is the one that planned the model.

Code	Information	Amendment
M3d	logit(M1) + name features centred within each AOI, without composition terms	2
M1co, M3o	as M1c and M3n with the HAND of the nearest settlement's site in place of the names	5
M1cL, M3oL	as M1c and M3n with the flooded share within 500 m of the nearest settlement in the same event (the cell itself left out)	6
M1cG, M3oG	as M1c and M3n with recurrent surface water within 500 m of the nearest settlement (occurrence and seasonality, 1984–2021)	6
M1cS, M3nS, M3nSp	as M1c, M3n and M3np with the strict reading (no springs or wells)	6
T0$^-$	T0 without river distances and without the water and wetland classes of land cover (rung of the ladder)	6
M1cEc, M3oEc	as M1c and M3n with the share of other EMS events (more than 30 days apart) in which the cell itself flooded, and whether there is such an event	7
M1cE, M3oE	the same share averaged over the cells within 500 m of each settlement and assigned through the nearest settlement	7
M1cF, M3nF, M3nFp	as M1c, M3n and M3np with the floodplain reading (riverine and wetland classes)	7
M1 + c(q), M1 + c(q) + q	reference and names models of a reassignment q, with the regional composition recomputed from the reassigned names	7
M1cW, M3oW	as M1c and M3n with recurrent surface water of the cell itself (occurrence, seasonality and maximum extent, 1984–2021, averaged over the cell)	8

Table A10: Post hoc checks of the within-area comparison (M1c to M3n or its analogue): R with its 90 % cluster-bootstrap interval and the verdict of the protocol's decision table. Amendment: the amendment that planned the check; code: the frozen code of the confirmatory run or the code with the corrected reading of Norwegian names. The measure of the baseline gain is that of each run (for the European AOIs, that of these AOIs).

Check	Amendment	Code	R	90 % interval	Verdict
Flood events held out whole	2	frozen	+0.018	[+0.001, +0.039]	not-meaning
Flood events held out whole	7	corrected	+0.004	[−0.004, +0.014]	equivalent-null
Flood traces counted as not flooded	5	corrected	−0.001	[−0.007, +0.004]	equivalent-null
Flooded share over land only	5	corrected	+0.003	[−0.002, +0.009]	equivalent-null
Partly flooded cells excluded	7	corrected	−0.004	[−0.010, +0.001]	equivalent-null
Every AOI weighted equally	6	frozen	−0.0003	[−0.004, +0.004]	—
European AOIs alone, own baseline gain	7	corrected	−0.0005	[−0.013, +0.013]	equivalent-null
Without the Americas and Oceania	5	corrected	−0.006	[−0.015, +0.002]	—
Strict reading	6	corrected	−0.003	[−0.006, +0.0004]	equivalent-null
Gazetteer cleaned of city sections and duplicates	6	corrected	−0.003	[−0.007, +0.0004]	equivalent-null
Without AOIs lacking a hydrography layer	6	corrected	−0.002	[−0.006, +0.004]	equivalent-null
Without AOIs lacking a flood type	6	corrected	−0.009	[−0.019, −0.001]	negative
Language-model reading, primary split	6	corrected	−0.001	[−0.008, +0.005]	equivalent-null
Floodplain reading (riverine and wetland classes)	7	corrected	−0.002	[−0.004, +0.0003]	equivalent-null
Cleaned lexicon (two-letter affixes and English suffixes removed)	8	corrected	−0.004	[−0.009, +0.001]	equivalent-null
GFM labels, existing predictions rescored	3	frozen	−0.003	[−0.020, +0.011]	equivalent-null

Table A11: The primary comparison in subsets of cells and under variants of the fold assignment (post hoc). For subsets, R is the names increment in the subset divided by the baseline gain in the same subset (as for the H5 strata); the 90 % interval comes from the cluster bootstrap over location groups restricted to the subset, and the last column gives the equivalence test alone. For variants, the last column gives the label of the pre-specified decision table and the equivalence test; the run with one seed instead of five is the pre-specified sensitivity analysis 5 (protocol §7-5). Built-up and tree cover removed: cells with ESA CCI built-up cover above 10 % or tree cover above 50 %. Distance bands up to 500 m, 0.5–1 km and beyond 1 km, the classes, the sensors, the fold reassignments and the activation grouping were planned in amendment 2 before they were run. The baseline gain within a class differs between classes by up to about seventeen-fold, so the verdicts by class depend strongly on their own denominators; Table A27 compares classes on the Brier scale. [a]Not planned (added while the analysis was written, or a subset of such an analysis). Values of R for subsets with a small baseline gain (for example single classes of names) are unstable; the increments by class on the Brier scale are in the table of increments by class.

Subset or variant	Cells	Units	Baseline ΔBS	R	90 % interval	Verdict
Distance from the cell to its nearest settlement						
up to 500 m	160,239	195	6.48×10^{-3}	−0.003	[−0.011, +0.004]	equivalent
up to 250 m[a]	44,030	195	6.65×10^{-3}	−0.001	[−0.008, +0.005]	equivalent
250–500 m[a]	116,209	195	6.41×10^{-3}	−0.004	[−0.012, +0.004]	equivalent
0.5–1 km	312,024	195	6.67×10^{-3}	−0.002	[−0.011, +0.006]	equivalent
more than 1 km	1,035,223	195	7.99×10^{-3}	−0.004	[−0.011, +0.002]	equivalent
up to 500 m, nearest name water-related[a]	10,082	134	5.84×10^{-3}	+0.060	[−0.036, +0.162]	not equivalent
up to 500 m, built-up and tree cover removed[a]	111,426	193	8.33×10^{-3}	−0.005	[−0.012, +0.003]	equivalent
up to 500 m, water-related, built-up and tree cover removed[a]	6,945	130	7.70×10^{-3}	+0.051	[−0.050, +0.155]	not equivalent
Class of the nearest name						
water-related (any)	89,404	146	1.25×10^{-2}	−0.004	[−0.023, +0.016]	equivalent
water	35,814	118	2.65×10^{-2}	+0.011	[−0.001, +0.025]	equivalent
wetland	2,857	51	1.54×10^{-3}	−0.499	[−1.262, −0.0002]	not equivalent
riverine	13,841	85	5.81×10^{-3}	−0.244	[−0.509, +0.004]	not equivalent
crossing	13,918	83	1.97×10^{-3}	+0.106	[−0.135, +0.395]	not equivalent
lowland	24,552	101	2.71×10^{-3}	+0.074	[−0.090, +0.265]	not equivalent
highland	58,021	130	8.62×10^{-3}	−0.013	[−0.038, +0.013]	equivalent
Post-event imagery of the delineation						
radar only	839,258	122	5.41×10^{-3}	−0.006	[−0.024, +0.008]	equivalent
optical only	174,932	37	2.19×10^{-2}	+0.0003	[−0.003, +0.004]	equivalent
not recorded (mostly schema generations 0–1)	493,296	65	6.10×10^{-3}	−0.004	[−0.012, +0.004]	equivalent
Fold assignment, grouping and seeds (frozen code; verdict of the decision table)						
pre-specified folds	1,507,486	195	7.55×10^{-3}	−0.003	[−0.011, +0.003]	equivalent-null (TOST holds)
folds reassigned (seed 1)	1,507,486	195	7.50×10^{-3}	−0.005	[−0.012, +0.0003]	equivalent-null (TOST holds)
folds reassigned (seed 2)	1,507,486	195	7.04×10^{-3}	+0.005	[−0.001, +0.013]	equivalent-null (TOST holds)
AOIs of the same activation grouped	1,507,486	106	9.53×10^{-3}	+0.018	[+0.001, +0.039]	not-meaning (TOST holds)
one seed instead of five	1,507,486	195	5.69×10^{-3}	−0.004	[−0.011, +0.002]	equivalent-null (TOST holds)

Table A12: The verdict in more detail (exploratory, amendment 8). Achieved bound: the smallest symmetric margin within which the 90 % interval of R lies. Location groups needed: the multiple of the present number of groups at which the 90 % interval would lie within ±0.05 (equivalence) or the lower 95 % limit would exceed zero (positive), if the half-width shrinks with the square root of the number of groups and the point estimate stays; — where the point estimate lies outside the margins. These multiples are power calculations at the observed point estimate and add nothing to the intervals; the paper no longer uses them. Permutations: 200 reassignments of names among the settlements of each AOI with the regional composition rebuilt. Refitted interval: 500 bootstrap replicates of location groups in which the second stage of both models is refitted with the replicate's weights, against the same replicates with the fitted models held fixed. Effect at the margin: an odds ratio of 2.34 for all ground nearest a water-related name injected into the labels, every model refitted, and each replicate judged by the protocol's decision table without the placebo condition.

Quantity	Value
Achieved equivalence bound, frozen code (verdict): R; Brier units	0.011; 8.3×10^{-5}
Achieved equivalence bound, corrected reading: R; Brier units	0.013; 9.5×10^{-5}
largest over five measures of the baseline gain and two bootstraps (frozen code)	0.046 (two-way, Y1)
Location groups needed, total increment: equivalence; positive (multiple of 195)	20.2; 11.3
Location groups needed, flexible models refitted: equivalence; positive (multiple of 195)	45.7; 7.0
Location groups needed, within 500 m of water-related names (not planned): equivalence; positive (multiple of 134)	—; 3.8
Permutations with rebuilt composition: one-sided p; mean and 95th percentile of R	0.72; −0.002, 0.004
label with the permutation distribution as the placebo	unchanged (equivalent-null)
Corrected run, second stage refitted in each replicate: 90 % (95 %) interval of R	−0.017 to 0.013 (−0.024 to 0.019)
Corrected run, same replicates, models fixed: 90 % (95 %) interval of R	−0.013 to 0.003 (−0.014 to 0.004)
standard deviation of R, refitted over fixed	2.31
Effect at the margin injected, every model refitted: replicates judged equivalent (Wilson 95 %)	0 of 40 (0.0 %–8.8 %)
labels; mean R	positive-small 40; 0.040

Table A13: The skill of the held-out predictions of the corrected run with and without the names, decomposed (exploratory; amendment 9). Murphy decomposition (Murphy, 1973) over ten bins of predicted probability (the binned form omits the within-bin variance and covariance terms, so reliability minus resolution plus uncertainty does not reproduce the Brier score exactly); within AOIs: each model's predictions shifted on the logit scale so that its mean in each AOI equals the observed flooded fraction (479 AOIs with both flooded and dry cells), and the comparison and measure of the baseline gain recomputed; maps: in each AOI, the cells with the highest predicted probability (for HAND, the lowest HAND, ties broken by mean HAND and then at random) flagged as flooded, as many as were observed flooded (479 AOIs). Differences of the decomposition and of the critical success index: M3n minus M1c; the names raised the index in 16 % of the AOIs and lowered it in 23 %. M1c: without names; M3n: with names; M1 ranks cells within AOIs as M1c does. Intervals: 95 % from a cluster bootstrap over location groups (4,000 replicates) unless stated.

Quantity	M1c	M3n	Difference or reference
Decomposition of the Brier score (Murphy, 1973)			
Brier score	0.04305	0.04308	3.1×10^{-5}
Reliability (lower is better)	2.0×10^{-4}	2.1×10^{-4}	8.0×10^{-6} (-5.7×10^{-6} to 3.6×10^{-5})
Resolution (higher is better)	8.0×10^{-3}	8.0×10^{-3}	-1.3×10^{-5} (-4.2×10^{-5} to 1.3×10^{-5})
Uncertainty	5.2×10^{-2}		
Within AOIs (each AOI's flooded fraction given)			
Baseline gain within AOIs	7.5×10^{-3}		
Names increment BS(M1c) − BS(M3n)			-3.1×10^{-5} (-5.8×10^{-5} to -7.2×10^{-6})
R (90 % interval)			−0.004 [−0.007, −0.001]
Maps flagging the observed number of flooded cells per AOI (median; quartiles)			
Critical success index	0.129 (0.03–0.27)	0.132 (0.03–0.27)	mean HAND: 0.084 (0.00–0.21)
Hit rate	0.229 (0.07–0.42)	0.233 (0.06–0.43)	mean HAND: 0.155
False-alarm ratio	0.771 (0.58–0.93)	0.767 (0.57–0.94)	mean HAND: 0.845
Difference in the critical success index			median 0.000 (0.000 to 0.000)

Table A14: Brier scores of the held-out predictions of each split (exploratory, amendment 10, item B9). Code: the frozen code of the confirmatory run or the corrected code; on the activation split the baseline gain is Y0 9.53×10^{-3}, Y1 3.94×10^{-3}, Y2 5.12×10^{-3}, Y3 4.75×10^{-3}, Y4 3.93×10^{-4}; Table S7 defines the measures of the baseline gain and the bootstraps. On the activation split the reference model M1c is worse than the recalibrated baseline model M1s. Lower block: R under each measure, with the 90 % interval of the cluster and the two-way bootstrap (†: the interval does not lie within ± 0.05).

Split	Code	M0	M1	M1s	M1c	M3n
Location groups (primary)	corrected	0.0513	0.0437	0.0434	0.0430	0.0431
Countries	frozen	0.0565	0.0494	0.0479	0.0485	0.0485
Time (from 2024)	frozen	0.0495	0.0472	0.0460	0.0468	0.0468
Language-model cut-off	corrected	0.0611	0.0499	0.0494	0.0499	0.0499
Activations kept whole	corrected	0.0583	0.0488	0.0477	0.0519	0.0518
Activations kept whole	frozen	0.0583	0.0488	0.0477	0.0518	0.0516
R on the activation split (corrected code) under each measure of the baseline gain: 90 % interval, cluster bootstrap and two-way bootstrap						
Y0: pre-specified			0.004	−0.004 to 0.014	−0.007 to 0.021	
Y1: lower 95 % bound			0.009	−0.008 to 0.033	−0.036 to 0.116†	
Y2: recalibrated			0.007	−0.007 to 0.025	−0.012 to 0.039	
Y3: from a constant			0.008	−0.007 to 0.027	−0.013 to 0.042	
Y4: stacked			0.091	−0.085 to 0.332†	−0.158 to 0.508†	

Table A15: Increment from the regional composition of names (AOI means of the name features; for the last column, of the language-model readings), as a fraction of the pre-specified baseline gain, with 95 % cluster-bootstrap intervals (descriptive). M0→M0c compares the flexible position model with a stacked model and therefore includes the gain from recalibration; M0s and T0s are the same base models recalibrated without the composition (post hoc, amendment 2), so M0s→M0c and T0s→T0c isolate the composition. [e]All areas of each flood event on the same side of every split (no T0 arm in this run). [u]The pre-specified rule leaves the baseline gain undefined on this split (its 95 % interval includes zero), so these fractions are only indicative. The lower block gives the composition of the language-model readings against each measure of the baseline gain of Table S7 (amendment 2, addendum 2): point estimate and 95 % interval under the bootstrap over location groups and under the two-way bootstrap over activations and countries; undefined where the measure of the baseline gain is not positive.

Split	M0→M0c	M0s→M0c	T0s→T0c	M1s→M1c	M1s→M1rc (LM)
Location groups	+0.416 [+0.005, +1.047]	+0.229 [−0.134, +0.726]	+0.045 [−0.273, +0.397]	+0.037 [−0.198, +0.330]	—
Flood events grouped[e]	+0.653 [+0.066, +1.473]	+0.071 [−0.561, +0.751]	—	−0.430 [−1.181, +0.126]	—
Countries	+0.578 [−0.128, +1.674]	+0.009 [−0.262, +0.232]	−0.125 [−0.373, +0.058]	−0.085 [−0.491, +0.189]	—
Time[u]	+0.531 [−1.364, +2.295]	−0.575 [−3.037, +0.863]	−2.240 [−6.859, +0.133]	−0.311 [−0.931, +0.413]	—
LM cut-off	+0.666 [−0.008, +1.507]	−0.226 [−0.679, +0.015]	−0.613 [−1.776, +0.007]	−0.030 [−0.087, +0.023]	+0.050 [−0.077, +0.189]
Composition of the language-model readings (M1s→M1rc, LM cut-off), by measure of the baseline gain					
	Y0	Y1	Y2	Y3	Y4
cluster	+0.050 [−0.077, +0.189]	+0.325 [−0.507, +1.239]	+0.328 [−0.512, +1.252]	+0.181 [−0.282, +0.689]	+0.143 [−0.222, +0.544]
two-way	+0.050 [−0.131, +0.163]	undefined	+0.328 [−0.871, +1.078]	+0.181 [−0.479, +0.593]	+0.143 [−0.378, +0.468]

Table A16: Whether areas with more water-related names flood more, within language regions (exploratory; amendment 10, item B13). Regression of the empirical logit of each AOI's flooded fraction, $\log\{(k+0.5)/(n-k+0.5)\}$, on the share of its analysed cells whose nearest name is water-related and on the AOI mean of the baseline model's held-out logit, with activation fixed effects, for the AOIs of countries read with the German, English or French lexicon blocks; AOIs that are the only analysed AOI of their activation drop out with the fixed effects. Standard errors clustered by activation (CR1), 95 % intervals from the t distribution with one degree of freedom fewer than the activations.

Language	AOIs	Activations	Share water-related	Mean logit of M1
German	54	16	−3.58 (−21.02 to +13.85)	+0.47 (+0.10 to +0.84)
English	111	30	−2.27 (−6.39 to +1.86)	+0.57 (+0.17 to +0.97)
French	58	14	−3.62 (−12.16 to +4.92)	+0.79 (−0.06 to +1.64)
All three	212	56	−2.38 (−6.04 to +1.28)	+0.56 (+0.29 to +0.84)

Table A17: Post hoc runs and controls of amendment 6, primary split, corrected code. R is the within-area increment as a fraction of the baseline gain of the same run, with the fixed-margin 90 % interval and the verdict of the decision table. At the T0 rung, the 95 % interval of R is −0.024 to 0.002 with the corrected code against −0.017 to 0.001 with the frozen code (Table 2 of the paper), so a 90 % interval here can be wider than the 95 % interval there. The spread of the location groups' contributions grew by a factor of 1.48, almost all of it from two large AOIs, EMSR097 AOI01 (BD) and EMSR838 AOI05 (PK), whose contributions changed when the models were refitted; without them the factor is 0.87.

Run	R	90 % interval	Verdict	Sample
Flood-label oracle (same event, leaks the outcome)	0.513	0.338 to 0.715	positive-meaningful	500 AOIs
Recurrent surface water (JRC GSW)	0.006	−0.009 to 0.022	equivalent-null	500 AOIs
Recurrent surface water without permanent water	0.004	−0.020 to 0.027	equivalent-null	500 AOIs
Names on T0$^-$ (no river distances, no water land cover)	−0.008	−0.019 to −0.001	negative	500 AOIs
Names on T0 (same run)	−0.008	−0.021 to 0.001	equivalent-null	500 AOIs
Strict reading of the names	−0.003	−0.006 to 0.0004	equivalent-null	500 AOIs
Language-model reading, primary split	−0.001	−0.008 to 0.005	equivalent-null	500 AOIs
Gazetteer without city sections, historical places, duplicates	−0.003	−0.007 to 0.0004	equivalent-null	500 AOIs
Without AOIs lacking a hydrography layer	−0.002	−0.006 to 0.004	equivalent-null	285 AOIs
Without AOIs lacking a flood type	−0.009	−0.019 to −0.001	negative	346 AOIs
Events held out whole: real increment among 200 reassignments	—	percentile 48	—	—
Every AOI weighted equally	−0.0003	−0.004 to 0.004	equivalent	—

What assignment through settlements can carry

Table A18: Injected odds ratio (shape A: all ground nearest a water-related name) whose mean simulated gain reaches each interval's upper limit of R minus the gain expected without an effect, under the four no-effect references (exploratory, amendment 13, item B2); in parentheses, the value on the scale of the estimator of Sect. 6.2 of the paper. A bound is not a test inversion, and the Monte Carlo ranges of the bound tables are not confidence intervals. References: recomputed composition (amendment 8), no reassignment, centred reassignments, and the reassignments of amendment 5.

Interval	Upper R	Recomputed	None	Centred	Amendment 5
Protocol's interval (models fixed)	0.003	1.25 (1.36)	1.17 (1.24)	1.28 (1.40)	1.47 (1.69)
Location groups, second stage refitted (amendment 8)	0.013	1.57 (1.83)	1.50 (1.73)	1.59 (1.86)	1.76 (2.11)
Location groups, refitted, denominator resampled (amendment 11)	0.008	1.40 (1.59)	1.34 (1.50)	1.43 (1.62)	1.64 (1.94)
Activations × countries, refitted (amendment 11)	0.036	2.11 (2.63)	2.06 (2.55)	2.13 (2.65)	2.25 (2.83)
Spread over fold assignments (amendment 5)	0.004	1.31 (1.45)	1.23 (1.34)	1.33 (1.47)	1.53 (1.77)
Corrected test (a)	0.004	1.29 (1.43)	1.21 (1.31)	1.31 (1.45)	1.51 (1.75)
Corrected test (b)	0.016	1.65 (1.95)	1.58 (1.85)	1.67 (1.97)	1.82 (2.20)
Corrected test (c)	0.031	1.99 (2.45)	1.93 (2.37)	2.00 (2.47)	2.16 (2.70)
Corrected test (d)	0.031	1.99 (2.45)	1.93 (2.37)	2.01 (2.48)	2.16 (2.70)
Effect within 1 km of the settlement (shape BK; amendment 12, item C1)					
Margin	0.050	7.55			
Models fixed		2.13	1.82	2.20	2.92
Second stage refitted		3.19	2.99	3.26	3.98

Table A19: Effects injected into the observed labels for all ground nearest a water-related name, with the base models fixed and with every model refitted on the injected labels (amendment 7; three repetitions with the injections of amendment 6): mean gain in ΔBS over the uninjected run ($\times 10^{-4}$), the same with fixed base models over the ten repetitions of amendment 6, the ratio of the refitted to the fixed gain, the baseline gain (M0 to M1) of the refitted models in the injected labels ($\times 10^{-4}$) and the share of held-out labels changed by the injection (%). Odds ratios at the margin and bounds are not read from this simulation: it has no effect below an odds ratio of 1.65, and a straight line from zero across the convex gain curve would understate them (amendment 8, addendum 9).

Odds ratio	Fixed	Fixed, 10 rep.	Refitted	Ratio	Baseline gain	Labels changed
1.65	1.36	1.37	1.23	0.91	78.9	0.15
2.72	5.12	5.18	4.88	0.95	78.9	0.34
4.48	12.38	12.39	11.66	0.94	78.1	0.55

Table A20: Odds ratios of flooding that the margin and the observed upper limits correspond to under the synthetic labels of the protocol's power analysis (amendment 2): an effect on all ground nearest a water-related name, and on the ground within 500 m of such a settlement. In parentheses the range over Monte Carlo resamples of the simulated curve; >: beyond the largest simulated effect (odds ratio 4.95 for all ground, 20.1 within 500 m). These translations assume the label model of the power analysis; the simulations on the observed labels (Table 4 of the paper) give lower values.

Target of the translation	All ground nearest the name	Within 500 m
Margin ($R = 0.05$)	4.12 (4.04–4.22)	> 20
Upper 90 % limit of the observed increment	1.53 (1.44–1.61)	6.94 (4.12–9.74)
Same, two-way bootstrap	1.72 (1.63–1.83)	9.52 (6.56–12.64)
Upper limit, confirmatory run	1.53	6.94
Upper limit, fold assignment 1	1.19	3.22
Upper limit, fold assignment 2	2.20	> 20
Upper limit, location groups formed with a 10 km buffer	1.76	10.14
Upper limit, events held out whole	4.10	> 20

Table A21: Positive control without flood labels (amendment 6): within-AOI difference between settlements whose name is in each class and other settlements in the mean occurrence of surface water (percentage points of time), its mean seasonality (months) and the occurrence without permanent water, within 500 m of the settlement point (JRC Global Surface Water 1984–2021); 95 % cluster-bootstrap intervals over location groups.

Names	Occurrence	Seasonality	Occurrence, non-permanent
water-related (any)	+1.05 (+0.34 to +2.00)	+0.12 (+0.04 to +0.23)	+0.60 (+0.13 to +1.25)
strict reading	+1.36 (+0.41 to +2.65)	+0.15 (+0.04 to +0.30)	+0.74 (+0.11 to +1.62)
water	+0.48 (-0.22 to +1.55)	+0.06 (-0.02 to +0.19)	+0.19 (-0.18 to +0.85)
wetland	-0.34 (-1.01 to +0.34)	-0.05 (-0.13 to +0.04)	-0.29 (-0.79 to -0.04)
riverine	+4.04 (+1.88 to +6.01)	+0.43 (+0.20 to +0.67)	+2.56 (+0.76 to +4.17)
crossing	+1.87 (+0.58 to +3.38)	+0.22 (+0.07 to +0.38)	+1.24 (+0.32 to +2.35)
lowland	+0.03 (-0.14 to +0.22)	+0.00 (-0.02 to +0.03)	-0.01 (-0.09 to +0.08)
highland	-0.47 (-0.79 to -0.14)	-0.06 (-0.11 to -0.02)	-0.28 (-0.47 to -0.10)

Flooding near settlements

Table A22: Positive control without flood labels (binary measures): within-AOI differences between cells (or settlements) whose nearest settlement name is in the class and other cells (or settlements) of the same AOI, with 95 % cluster-bootstrap intervals over location groups. "HAND ≤ 2 m" is whether any 25 m pixel of the 250 m cell lies within 2 m of the nearest drainage (the cell's minimum HAND); it holds for most cells, so the shares in each group are given. Lower half: the cell lies in the lower half of its AOI by mean elevation. Continuous measures are in Table A23. The all-cell rows for water-related names as a group and for highland names are the analysis planned in amendment 2; the other all-cell rows (the split by class) were computed with it and are not planned. [a]Not planned: cells within 177 m of the settlement point (the one to four cells around it), averaged per settlement (18,558 settlements; AOIs need at least two settlements of each kind). [b]Excluding settlements whose GeoNames coordinates fall on whole arc-minutes (rounding of up to about 0.9 km in each coordinate; 11 % of the settlement points of this analysis, against 14 % of the settlements nearest the analysed cells).

Nearest name	Scale	HAND ≤ 2 m in cell (pp)	Class/other (%)	Lower half (pp)	AOIs
water-related (any)	all cells	+0.2 [-1.0, +1.3]	82/82	+0.6 [-2.9, +4.1]	310
	at settlement[a]	+5.7 [+3.5, +8.1]	89/83	+5.0 [+1.5, +8.7]	151
water	all cells	-0.7 [-2.0, +0.4]	83/83	-4.0 [-8.9, +0.7]	176
	at settlement[a]	+3.2 [+0.1, +6.6]	88/84	-3.7 [-9.7, +2.7]	64
wetland	all cells	+0.4 [-3.3, +3.8]	85/84	+1.5 [-7.4, +11.3]	41
	at settlement[a]	+6.7 [-4.3, +22.1]	88/81	-1.5 [-18.9, +25.0]	12
riverine	all cells	+2.1 [+0.1, +4.4]	90/87	+12.8 [+5.1, +20.3]	95
	at settlement[a]	+4.7 [+0.5, +10.4]	92/87	+16.4 [+8.9, +24.5]	38
crossing	all cells	+2.1 [+0.6, +3.6]	81/79	+9.4 [+4.3, +14.5]	102
	at settlement[a]	+7.5 [+0.5, +14.5]	91/84	+9.3 [+0.8, +17.6]	34
lowland	all cells	-0.9 [-3.2, +1.2]	78/79	-4.9 [-10.4, +0.9]	160
	at settlement[a]	+5.5 [+0.4, +10.7]	85/79	+3.4 [-3.7, +9.8]	60
highland	all cells	-3.1 [-4.5, -1.6]	78/81	-7.7 [-11.7, -3.5]	253
	at settlement[a]	-10.1 [-13.4, -6.8]	72/82	-8.9 [-13.0, -4.9]	137
water-related (any)	at settlement, coordinates not rounded[b]	+6.1 [+3.7, +8.4]	90/83	+4.9 [+1.3, +8.7]	135
highland	at settlement, coordinates not rounded[b]	-10.4 [-13.9, -6.8]	71/82	-9.0 [-13.2, -4.9]	122

Table A23: Positive control without flood labels (continuous measures), as in Table A22: differences in log(1 + distance to the nearest mapped river), in the cell's mean height above nearest drainage and in its HAND percentile within the AOI. The all-cell rows for water-related names as a group and for highland names are the analysis planned in amendment 2; the other all-cell rows (the split by class) were computed with it and are not planned. [a]Not planned: cells within 177 m of the settlement point (the one to four cells around it), averaged per settlement (18,558 settlements; AOIs need at least two settlements of each kind). [b]Excluding settlements whose GeoNames coordinates fall on whole arc-minutes (rounding of up to about 0.9 km in each coordinate; 11 % of the settlement points of this analysis, against 14 % of the settlements nearest the analysed cells).

Nearest name	Scale	log(1 + river distance)	Mean HAND (m)	HAND percentile (pp)
water-related (any)	all cells	-0.04 [-0.09, +0.01]	+0.7 [-0.7, +2.3]	-0.04 [-1.6, +1.7]
	at settlement[a]	-0.22 [-0.33, -0.11]	-2.0 [-2.9, -1.3]	-4.6 [-6.1, -3.1]
water	all cells	+0.01 [-0.07, +0.08]	+1.1 [-0.3, +2.8]	+2.0 [-0.1, +4.4]
	at settlement[a]	-0.17 [-0.33, -0.01]	-0.6 [-1.7, +0.5]	-2.0 [-4.3, +0.4]
wetland	all cells	-0.001 [-0.14, +0.14]	+0.1 [-2.8, +3.7]	-3.5 [-8.2, +1.6]
	at settlement[a]	+0.17 [-0.28, +0.50]	-0.7 [-5.7, +3.9]	-1.7 [-13.6, +5.3]
riverine	all cells	-0.01 [-0.15, +0.16]	-0.6 [-3.1, +2.2]	-5.0 [-8.7, -1.3]
	at settlement[a]	-0.12 [-0.42, +0.14]	-2.2 [-4.1, -0.7]	-7.7 [-11.9, -4.4]
crossing	all cells	-0.18 [-0.27, -0.10]	-1.0 [-3.1, +0.8]	-1.5 [-4.2, +0.9]
	at settlement[a]	-0.54 [-0.86, -0.28]	-2.6 [-4.5, -1.0]	-5.3 [-9.9, -1.5]
lowland	all cells	-0.02 [-0.11, +0.07]	+1.7 [-1.0, +5.0]	+1.1 [-1.6, +3.9]
	at settlement[a]	-0.22 [-0.40, -0.05]	-2.3 [-3.9, -0.9]	-3.2 [-6.7, +0.1]
highland	all cells	+0.08 [+0.03, +0.13]	+3.4 [+1.5, +5.9]	+3.8 [+2.1, +5.5]
	at settlement[a]	+0.11 [+0.01, +0.23]	+5.3 [+3.8, +7.1]	+8.5 [+6.0, +10.8]
water-related (any)	at settlement, coordinates not rounded[b]	-0.23 [-0.35, -0.11]	-2.1 [-2.9, -1.5]	-5.0 [-6.7, -3.5]
highland	at settlement, coordinates not rounded[b]	+0.11 [-0.01, +0.22]	+5.3 [+3.7, +7.1]	+8.8 [+6.3, +11.3]

Table A24: Where settlements with water-related names stand (exploratory, amendment 8; no flood labels). Within-AOI difference between settlements whose name is in the class and other settlements of the same AOI in the height of the settlement's cell above the nearest point of a river draining at least 100 or 1000 km^2 (HydroRIVERS; mean elevation of the settlement's cell minus the lowest elevation of the river's cell, m), and in the logarithm of the drainage area of the nearest river; 95 % cluster-bootstrap intervals over location groups. Last row: the HAND of the settlement's site, for names that contain the name of a river, stream or lake within 5 km in the gazetteer.

Names	Above rivers ≥100 km^2	≥1000 km^2	log drainage area	Settlements
Water-related (pooled)	−5.0 (−9.0 to −1.3)	−4.8 (−9.0 to −1.0)	0.11 (−0.02 to 0.24)	11120
Floodplain reading (riverine, wetland)	−4.9 (−10.6 to −0.2)	−3.2 (−9.6 to 2.6)	0.28 (−0.09 to 0.67)	5150
Water bodies	0.9 (−4.5 to 6.6)	−0.9 (−6.8 to 5.1)	−0.21 (−0.48 to 0.05)	6156
Wetland	0.1 (−17.3 to 13.5)	5.3 (−17.4 to 28.2)	0.18 (−0.90 to 1.39)	1478
Riverine ground	−4.4 (−12.0 to 2.3)	−5.7 (−13.1 to 0.9)	0.31 (−0.07 to 0.71)	4077
Crossings	−8.7 (−13.9 to −3.9)	−7.5 (−13.2 to −3.0)	0.61 (0.18 to 1.13)	3121
Lowland	−6.0 (−17.3 to 4.7)	−4.3 (−16.8 to 7.4)	0.21 (−0.03 to 0.43)	5565
Names containing a nearby river or lake name (HAND of the site)	−2.4 (−4.4 to −0.6)	—	—	3931

Table A25: Within-AOI odds ratios of flooding for cells whose nearest settlement name is in the class, from logistic regressions with AOI fixed effects, unadjusted or with the logit of the held-out terrain-model prediction (M1) as a covariate; 95 % intervals from cluster-robust (location-group) standard errors; 479 AOIs (AOIs whose sampled cells are all dry carry no information and are omitted). [r] All cells, one indicator per class: the analysis planned in amendment 2. The distance-banded models (class × band, with band main effects) were not planned. The distance-band row gives the odds of a flood label 0.5–1 km and more than 1 km from the nearest settlement relative to within 500 m. Built-up and tree cover removed: cells with ESA CCI built-up cover above 10 % or tree cover above 50 % excluded. A cluster bootstrap over location groups (200 replicates, AOIs re-entered with their own intercepts) gives similar intervals, e.g. 0.73–1.28 for water-related names within 500 m given the baseline model.

Nearest name	All cells[r]		Unadjusted, by distance			Given the baseline model, by distance		
	unadjusted	given terrain	≤ 500 m	0.5–1 km	> 1 km	≤ 500 m	0.5–1 km	> 1 km
water-related (any)	1.00 [0.81, 1.23]	1.02 [0.83, 1.25]	1.05 [0.80, 1.36]	1.11 [0.91, 1.34]	0.95 [0.74, 1.22]	1.01 [0.77, 1.32]	1.06 [0.85, 1.33]	1.00 [0.80, 1.26]
water	0.85 [0.60, 1.21]	0.97 [0.73, 1.29]	0.89 [0.67, 1.20]	0.86 [0.67, 1.11]	0.85 [0.52, 1.38]	0.77 [0.57, 1.05]	0.80 [0.62, 1.03]	1.07 [0.74, 1.55]
wetland	0.98 [0.46, 2.09]	1.09 [0.47, 2.53]	1.19 [0.52, 2.73]	1.10 [0.45, 2.69]	0.92 [0.45, 1.84]	1.60 [0.71, 3.59]	1.32 [0.48, 3.62]	0.80 [0.38, 1.67]
riverine	1.30 [0.91, 1.86]	1.17 [0.74, 1.83]	1.46 [0.94, 2.26]	1.48 [1.03, 2.11]	1.16 [0.80, 1.69]	1.41 [0.92, 2.16]	1.38 [0.92, 2.08]	1.04 [0.64, 1.70]
crossing	1.02 [0.72, 1.44]	0.90 [0.65, 1.26]	0.75 [0.42, 1.35]	1.01 [0.64, 1.60]	1.04 [0.71, 1.53]	0.77 [0.46, 1.26]	0.89 [0.58, 1.35]	0.92 [0.63, 1.35]
lowland	0.83 [0.56, 1.23]	0.93 [0.65, 1.33]	0.80 [0.42, 1.54]	1.06 [0.61, 1.86]	0.76 [0.49, 1.18]	0.87 [0.50, 1.50]	1.16 [0.68, 1.97]	0.87 [0.59, 1.29]
highland	1.00 [0.83, 1.22]	1.08 [0.90, 1.30]	0.98 [0.74, 1.29]	1.04 [0.80, 1.35]	1.00 [0.81, 1.24]	1.21 [0.88, 1.68]	1.10 [0.89, 1.36]	1.07 [0.86, 1.33]
Distance band (all names)	—	—	1 (reference)	1.61 [1.48, 1.76]	1.97 [1.64, 2.37]	1 (reference)	1.12 [1.05, 1.20]	1.29 [1.12, 1.48]
water-related, built-up and tree cover removed	—	—	1.03 [0.79, 1.36]	1.09 [0.90, 1.34]	0.94 [0.73, 1.22]	1.03 [0.77, 1.37]	1.09 [0.86, 1.37]	1.01 [0.79, 1.29]

Table A26: Odds ratios of flooding near settlements by class and distance band on the EMS labels, given the baseline model's prediction (AOI fixed effects), with cluster-robust and pairs cluster bootstrap 95 % intervals over location groups (2,000 replicates; amendment 7). The two blocks are separate models (water-related names; the six classes).

Names, distance	Odds ratio	Cluster-robust	Pairs bootstrap
water-related, ≤500 m	1.01	0.77–1.32	0.77–1.31
water-related, 0.5–1 km	1.06	0.85–1.33	0.84–1.30
water-related, >1 km	1.00	0.80–1.26	0.79–1.25
water, ≤500 m	0.77	0.57–1.05	0.54–1.03
water, 0.5–1 km	0.80	0.62–1.03	0.60–1.02
water, >1 km	1.07	0.74–1.55	0.71–1.49
wetland, ≤500 m	1.60	0.71–3.59	0.41–3.62
wetland, 0.5–1 km	1.32	0.48–3.62	0.32–3.06
wetland, >1 km	0.80	0.38–1.67	0.35–1.61
riverine, ≤500 m	1.41	0.92–2.16	0.81–2.19
riverine, 0.5–1 km	1.38	0.92–2.08	0.84–2.04
riverine, >1 km	1.04	0.64–1.70	0.60–1.58
crossing, ≤500 m	0.77	0.46–1.26	0.42–1.22
crossing, 0.5–1 km	0.89	0.58–1.35	0.56–1.32
crossing, >1 km	0.92	0.63–1.35	0.59–1.29
lowland, ≤500 m	0.87	0.50–1.50	0.45–1.41
lowland, 0.5–1 km	1.16	0.68–1.97	0.61–1.83
lowland, >1 km	0.87	0.59–1.29	0.57–1.23
highland, ≤500 m	1.21	0.88–1.68	0.83–1.64
highland, 0.5–1 km	1.10	0.89–1.36	0.85–1.34
highland, >1 km	1.07	0.86–1.33	0.85–1.35

Table A27: The names increment within areas (M1c→M3n) by class of the nearest name, on the Brier scale (post hoc; planned in amendment 2 as descriptive quantities, the 95 % intervals and the difference between real and permuted names in its fourth addendum). All Brier-score differences are in units of 10^{-4}; positive values mean that names improved held-out predictions. Intervals come from the cluster bootstrap over location groups restricted to the cells of each class (classes overlap: a name can belong to several). Baseline ΔBS is the baseline gain (M0→M1) in the same cells, the denominator of the class verdicts in Table A11; the pooled equivalence margin is 3.78. [a]For all cells, the difference between real and permuted names is given as a ratio in Table S6.

Nearest name	Cells	Units	ΔBS	90 % interval	95 % interval	Real − permuted (95 %)	Baseline ΔBS
all cells (primary)	1,507,486	195	−0.264	[−0.826, +0.210]	[−0.956, +0.279]	—[a]	75.5
water-related (any)	89,404	146	−0.448	[−2.87, +1.98]	[−3.30, +2.39]	+0.895 [−3.16, +5.23]	125
water	35,814	118	+2.84	[−0.263, +6.55]	[−0.777, +7.51]	+6.33 [+0.124, +15.2]	265
wetland	2,857	51	−7.68	[−19.4, −0.00298]	[−22.6, +0.420]	−10.6 [−24.7, −1.26]	15.4
riverine	13,841	85	−14.2	[−29.6, +0.259]	[−33.2, +2.93]	−11.0 [−31.1, +8.00]	58.1
crossing	13,918	83	+2.08	[−2.65, +7.77]	[−3.61, +9.03]	+0.585 [−5.71, +8.62]	19.7
lowland	24,552	101	+2.01	[−2.43, +7.18]	[−3.26, +8.38]	+1.15 [−4.03, +6.98]	27.1
highland	58,021	130	−1.10	[−3.26, +1.12]	[−3.63, +1.54]	−0.761 [−2.79, +1.45]	86.2

Table A28: Within-AOI flood contrasts by distance from the cell to its nearest settlement (not planned; added while the planned analyses were written, with their full reporting planned in amendment 2, addendum 2): the share of cells labelled flooded nearest a water-related name minus the share nearest other names, in percentage points, observed and net of the held-out predictions of the baseline model with the regional composition of names (M1c), with 95 % cluster-bootstrap intervals over location groups. AOIs enter a band when both groups have at least 20 cells in it; they are averaged with weights equal to the size of the smaller group, or scaled to the uncapped number of evaluable cells of each AOI, which lets a few large AOIs dominate.

Distance to the nearest settlement	AOIs	Location groups	Weights: smaller group		Weights: uncapped AOI size	
			observed	net of terrain	observed	net of terrain
≤ 250 m	37	31	+1.1 [+0.04, +2.9]	+0.5 [−0.3, +1.8]	+0.1 [−1.0, +2.4]	−0.8 [−2.3, +1.6]
250–500 m	98	70	+0.2 [−0.6, +1.2]	−0.1 [−0.8, +0.9]	−1.1 [−2.6, +0.9]	−2.1 [−4.3, +0.6]
0.5–1 km	185	111	+0.4 [−0.4, +1.4]	+0.2 [−0.5, +1.1]	+0.8 [−1.7, +3.9]	−0.04 [−2.8, +2.9]
> 1 km	263	134	−0.2 [−1.1, +0.7]	−0.1 [−0.9, +0.6]	+2.9 [−0.4, +5.5]	+1.6 [−0.7, +3.4]

Table A29: Odds ratios of flooding within 500 m of settlements, given the held-out terrain prediction (AOI fixed effects, distance bands, logit of the baseline model's prediction; exploratory, amendment 8, all not planned in the protocol). Pairs: cluster bootstrap over location groups with the regression refitted, 2,000 replicates. Cleaned reading: without two-letter affixes and the English suffixes that also read transferred names (0.6 % of the cells change their reading). Uncapped: every evaluable cell of the analysed AOIs, with the predictions of the uncapped sensitivity run. Misclassification: probabilistic bias analysis with the rates measured against recorded etymologies by the model that compiled the lexicon. [s]AOIs that enter the model, which needs flooded and dry cells in an AOI, of the AOIs whose EMS map was not drawn from Sentinel-1 (capped sample; other tables count the AOIs of their own samples). Columns 4–6: flooded cells within 500 m of water-related names, location groups with any cell within 500 m of a water-related name (flooded or not), and the AOIs that enter each model (AOIs with flooded and dry cells). The uncapped row uses the readings of the frozen code; the corrected readings are in the table of uncapped estimates of amendment 11.

Estimate	Odds ratio	95 % interval	Flooded	Groups	AOIs
Pooled water-related class, EMS labels (pairs cluster bootstrap)					
All AOIs	1.01	0.77–1.31	361	134	479
AOIs with a recorded flood type	1.00	0.77–1.30	214	91	331
Without AOIs lacking a hydrography layer	0.85	0.62–1.19	183	84	269
Without Oceania and the Americas	0.96	0.71–1.29	303	107	402
Without Australia	1.00	0.74–1.30	322	124	436
Cleaned reading	1.04	0.76–1.38	271	140	479
Every evaluable cell (no cap; frozen-code readings)	0.91	0.76–1.15	3,749	137	500
Without AOIs in Pakistan	1.09	0.77–1.45	271	127	467
Without AOIs in Pakistan and Bangladesh	0.97	0.72–1.24	184	125	465
Truly water-related names (misclassification corrected)	1.00	0.63–1.63			
By the river nearest the settlement, and adjusted for its height (cluster-robust)					
Draining less than 100 km^2	0.91	0.58–1.42	188	117	431
Draining 100–1000 km^2	1.61	0.98–2.64	87	93	351
Draining at least 1000 km^2	0.84	0.52–1.36	86	89	368
Cells with a height above the main river, unadjusted	1.00	0.75–1.33	331	118	427
adjusted for the height above the river	1.00	0.75–1.33	331	118	427
Names containing a nearby river or lake name	0.88	0.58–1.33	66	112	479
By class, all classes in one model (Bonferroni 95 % interval; Holm-adjusted p in the last column)					
Water bodies	0.77	0.51–1.17			0.58
Wetland	1.60	0.54–4.76			0.94
Riverine ground	1.41	0.80–2.50			0.58
Crossings	0.77	0.39–1.50			0.94
Lowland	0.87	0.42–1.81			0.94
Highland	1.21	0.79–1.87			0.94
Riverine flood cover as a fractional outcome, water-related class by distance band					
Within 500 m	0.98	0.80–1.21			
500 m–1 km	1.05	0.86–1.27			
Beyond 1 km	1.01	0.83–1.23			
The two label sets on the same cells (pairs bootstrap; jackknife with t quantiles)					
EMS	0.78	0.60–1.07	80 groups, 176 AOIs; jackknife 0.54–1.11		
GFM	0.55	0.34–0.81	80 groups, 176 AOIs; jackknife 0.36–0.85		
Ratio of the odds ratios, GFM over EMS	0.71	0.47–0.92			
Floodplain reading, EMS	0.83	0.38–1.81	14	31	182
Floodplain reading, GFM	0.31	0.09–1.00	5	31	182
EMS map not from Sentinel-1: GFM	0.48	0.24–0.98			63 of 68[s]
EMS map not from Sentinel-1: EMS	0.77	0.46–1.28			66 of 68[s]

Table A30: Where the flooded cells within 500 m of settlements with water-related names lie (EMS labels; descriptive): 361 flooded cells among 10,048 such cells, in the AOIs of the odds ratios near settlements (AOIs with both flooded and dry cells). Share: of all flooded cells near water-related names. For the AOIs in Pakistan the fourth column gives the flooded cells of all evaluable cells (amendment 8, addendum 6). AOIs: the AOIs with such cells whose nearest settlement lies in the country or activation; an AOI whose cells are nearest settlements in several countries counts in each, so the column sums to more than the number of AOIs (510 by country).

Country or activation	Flooded	Cells	Share	AOIs
By country of the nearest settlement				
PK	90	683	25 %	12
BD	87	453	24 %	2
AU	39	728	11 %	43
DE	37	1,888	10 %	29
PT	23	315	6.4 %	9
GB	15	1,807	4.2 %	18
MX	11	224	3.0 %	3
SI	8	180	2.2 %	6
VN	6	278	1.7 %	16
ES	6	142	1.7 %	25
MM	5	192	1.4 %	11
IR	4	152	1.1 %	3
RO	3	227	0.8 %	12
HR	3	101	0.8 %	27
KH	3	37	0.8 %	4
Other 30 countries	21	2,641	5.8 %	290
The activations with the most such cells				
EMSR097: Floods in Bangladesh (2014)	87	453	24 %	2
EMSR629: Floods in Pakistan (2022)	73	149	20 %	3
EMSR047: Flood in Sachsen-Anhalt- Germany (2013)	23	199	6.4 %	2
EMSR567: Floods in Queensland- Australia (2022)	22	159	6.1 %	8
EMSR864: Flood in Portugal and Spain (2026)	22	124	6.1 %	5
EMSR838: Flood in Pakistan (2025)	17	525	4.7 %	7
EMSR570: Flood in New South Wales- Australia (2022)	14	47	3.9 %	2
EMSR845: Flood in Mexico (2025)	9	88	2.5 %	1
EMSR150: Flood in England (2015)	6	470	1.7 %	4
EMSR388: Flood in the Southeast of Spain (2019)	6	62	1.7 %	1
AOIs in Pakistan, by activation (evaluable cells; flooded near / near)				
EMSR838: Flood in Pakistan (2025)	17	525	6,047 of 38,140	7
EMSR629: Floods in Pakistan (2022)	73	149	12,068 of 20,000	3
EMSR631: Flood in Pakistan (2022)	0	9	1,165 of 3,122	2

Table A31: The pooled odds ratio of flooding within 500 m of water-related names (EMS labels, the main model of amendment 8) with each activation that has flooded cells near water-related names left out in turn (exploratory; amendment 9; 48 activations). Flooded: flooded cells near water-related names in the activation left out. Intervals: 95 %, cluster-robust over location groups.

Left out	Flooded	OR	95 %	Left out	Flooded	OR	95 %
EMSR044	2	1.01	0.77–1.33	EMSR427	2	1.00	0.77–1.31
EMSR047	23	1.00	0.76–1.32	EMSR429	2	1.01	0.77–1.32
EMSR052	2	1.02	0.78–1.33	EMSR514	3	1.01	0.77–1.32
EMSR055	1	1.01	0.77–1.32	EMSR517	2	1.01	0.77–1.32
EMSR069	4	1.03	0.78–1.35	EMSR567	22	0.98	0.74–1.28
EMSR070	4	1.01	0.77–1.32	EMSR570	14	1.01	0.77–1.34
EMSR073	1	1.01	0.77–1.32	EMSR574	3	1.01	0.77–1.32
EMSR097	87	0.91	0.72–1.14	EMSR629	73	1.08	0.81–1.44
EMSR107	4	1.00	0.76–1.31	EMSR664	1	1.01	0.77–1.32
EMSR130	5	1.04	0.80–1.35	EMSR708	1	1.01	0.77–1.32
EMSR142	1	1.01	0.77–1.32	EMSR712	4	1.01	0.77–1.32
EMSR147	3	1.01	0.77–1.33	EMSR722	5	1.01	0.77–1.32
EMSR150	6	1.00	0.76–1.31	EMSR754	3	1.02	0.78–1.34
EMSR167	1	1.01	0.77–1.32	EMSR761	1	1.01	0.77–1.32
EMSR199	1	1.00	0.77–1.32	EMSR774	2	1.01	0.77–1.32
EMSR203	2	1.01	0.77–1.32	EMSR799	2	1.01	0.77–1.32
EMSR258	1	1.01	0.77–1.32	EMSR802	2	1.01	0.77–1.32
EMSR275	1	1.01	0.77–1.32	EMSR838	17	1.01	0.76–1.34
EMSR293	1	1.01	0.77–1.32	EMSR845	9	0.98	0.75–1.29
EMSR352	1	1.02	0.78–1.33	EMSR848	3	1.02	0.78–1.34
EMSR388	6	0.99	0.76–1.30	EMSR857	1	1.01	0.77–1.32
EMSR407	2	1.00	0.77–1.31	EMSR861	1	1.01	0.77–1.32
EMSR419	3	1.00	0.77–1.32	EMSR864	22	0.97	0.74–1.28
EMSR424	3	1.01	0.77–1.33	EMSR866	1	1.01	0.77–1.32

Table A32: The six class coefficients of the model with all lexicon classes within 500 m (EMS labels; the figure of odds ratios near settlements in the paper): unadjusted p, Holm-adjusted p (Holm, 1979) and Benjamini–Hochberg q (Benjamini and Hochberg, 1995) (exploratory; amendment 9).

Class	p	Holm p	BH q
Water bodies	0.097	0.584	0.339
Wetland	0.258	0.942	0.356
Riverine ground	0.113	0.584	0.339
Crossings	0.297	0.942	0.356
Lowland	0.609	0.942	0.609
Highland	0.236	0.942	0.356

Table A33: Odds ratio near water-related names corrected for misreading over a grid of rates (exploratory, amendment 10, item B5): $\mathrm{OR} = (\mathrm{OR}^*(1-p_0)-(1-p_1))/(p_1-\mathrm{OR}^*p_0)$, with OR^* the observed estimate within 500 m and the limits of its pairs-bootstrap 95 % interval; p_1: share of water-related names among those read as water-related; p_0: share among the others. Undefined: the observed value cannot arise from these rates.

p_1	p_0	Observed point	Observed lower	Observed upper
0.6	0.05	1.02	0.60	1.57
0.6	0.10	1.02	0.57	1.64
0.6	0.20	1.02	0.49	1.89
0.6	0.30	1.03	0.39	2.45
0.7	0.05	1.01	0.66	1.48
0.7	0.10	1.02	0.64	1.53
0.7	0.20	1.02	0.59	1.69
0.7	0.30	1.02	0.52	1.98
0.8	0.05	1.01	0.70	1.41
0.8	0.10	1.01	0.69	1.45
0.8	0.20	1.02	0.65	1.56
0.8	0.30	1.02	0.60	1.74
0.9	0.05	1.01	0.74	1.36
0.9	0.10	1.01	0.73	1.39
0.9	0.20	1.01	0.70	1.47
0.9	0.30	1.02	0.66	1.59

Table A34: Odds ratio of flooding within 500 m of settlements with water-related names against other names of the same AOI, given the held-out prediction of the baseline model (exploratory; amendment 11, items B2 and B5): AOI fixed effects, distance bands and the logit of the prediction, with a 95 % pairs cluster bootstrap over location groups (2,000 replicates). Flooded: flooded cells within 500 m of water-related names and the location groups that hold them. MDE: the smallest odds ratios below and above one detectable with 80 % power (two-sided 5 %). The uncapped sample conditions on the predictions of the uncapped sensitivity run (the baseline model does not use names). The GFM labels are not independent of EMS where the EMS map was drawn from Sentinel-1. The number of AOIs not mapped from Sentinel-1 differs between tables and figures because each counts the AOIs that enter its own sample: uncapped or capped, cells that GFM can assess, and AOIs with both flooded and dry cells.

Sample and labels	Odds ratio	95 % interval	Flooded	Groups	MDE
Every evaluable cell (no cap), corrected-code readings					
EMS labels, 500 AOIs	0.91	0.76–1.16	3,753	75	0.74, 1.35
GFM labels, cells that GFM can assess	0.82	0.60–1.07	434	29	0.66, 1.50
EMS labels on the same cells	0.89	0.67–1.16	560	36	0.68, 1.47
GFM labels, AOIs not mapped from Sentinel-1 (111)	0.85	0.43–1.46	166	15	0.44, 2.29
EMS labels, the same AOIs and cells	1.00	0.71–1.58	263	20	0.56, 1.79
EMS labels, flooded fraction over observed non-permanent water	0.91	0.77–1.17	3,810	76	0.74, 1.35
Capped table (10,000 cells per location group)					
EMS labels	1.01	0.77–1.31	361	52	—
EMS labels, flooded fraction over observed non-permanent water	1.02	0.78–1.32	372	54	0.68, 1.46

Table A35: The floodplain reading (riverine and wetland classes) within 500 m of settlements, given the terrain prediction (exploratory, amendment 10, item B2; defined after the class estimates had been seen). Intervals: pairs cluster bootstrap over location groups, 2,000 replicates. Countries are those of the nearest settlement. ^{x}Forms removed in the re-reading. Columns 4–5: flooded cells and location groups with such cells (or cells, for the forms).

Estimate	Odds ratio	95 % interval	Flooded	Groups
All AOIs (floodplain reading)	1.45	0.93–2.08	163	85
re-read through the same code, all forms	1.42	0.90–2.02	163	91
re-read without bank, au, dike and polder forms	1.54	0.94–2.28	142	85
Without activation EMSR097	1.39	0.75–2.28	82	83
One country left out				
without BD (81 flooded cells)	1.39	0.75–2.23	82	83
without DE (22 flooded cells)	1.55	0.92–2.31	141	79
without AU (16 flooded cells)	1.35	0.83–1.84	147	77
without MX (9 flooded cells)	1.38	0.86–2.00	154	83
without PK (8 flooded cells)	1.55	0.98–2.25	155	80
without SI (8 flooded cells)	1.40	0.87–1.99	155	83
without ES (6 flooded cells)	1.40	0.86–2.03	157	80
without GB (3 flooded cells)	1.46	0.90–2.10	160	77
without RO (3 flooded cells)	1.44	0.88–2.10	160	82
without BE (2 flooded cells)	1.44	0.91–2.07	161	85
without KH (2 flooded cells)	1.45	0.92–2.08	161	85
without AL (1 flooded cell)	1.44	0.90–2.05	162	84
without MM (1 flooded cell)	1.45	0.93–2.05	162	84
without VN (1 flooded cell)	1.48	0.93–2.12	162	79
Composition by matched form (flooded cells within 500 m; top 12)				
riverine *char*			71	340 cells
riverine *au*^{x}			19	459 cells
riverine *island*			14	27 cells
wet *bil*			9	59 cells
wet *estero*			9	18 cells
riverine *otok*			8	34 cells
riverine *arenal*			6	19 cells
wet *kallar*			5	7 cells
riverine *lunca*			3	74 cells
riverine *bela*			3	23 cells
riverine *kaoh*			2	10 cells
riverine *flat*			2	49 cells

Table A36: Independent labels from GFM (post hoc, amendment 3) on the 192 AOIs assessable on both label sets. Odds ratios come from logistic regressions with an intercept for each AOI and the logit of the held-out terrain prediction of the run with the frozen code, with standard errors clustered by location group; pct. points are within-AOI contrasts. The EMS odds ratio within 500 m of water-related names on these AOIs differs in the second decimal with the terrain prediction it is conditioned on: 0.79 here, 0.78 with the prediction of the run with the corrected reading (the paper and the table of the pooled odds ratio near settlements) and 0.80 with the baseline model refitted on GFM labels. [a]Added after the result above had been seen (amendment 3, addendum 2). [b]90 % interval.

Quantity	Labels	Estimate	95 % interval	Note
Agreement, all AOIs	both	93.2 % ($\kappa = 0.64$)	—	234,703 cells, 192 AOIs
Agreement, EMS image includes Sentinel-1	both	95.0 % ($\kappa = 0.69$)	—	142,614 cells, 124 AOIs
Agreement, EMS image from other radar	both	96.0 % ($\kappa = 0.71$)	—	50,892 cells, 50 AOIs
Agreement, EMS image optical	both	83.8 % ($\kappa = 0.50$)	—	41,197 cells, 18 AOIs
EMS − GFM, ≤ 500 m minus > 1 km (pct. points)	both	+1.0	+0.2 to +1.9	133 AOIs
EMS − GFM, water-related minus other names, ≤ 500 m	both	+0.4	−0.1 to +1.1	21 AOIs
EMS − GFM, water-related minus other names, all cells	both	−0.3	−1.0 to +0.6	82 AOIs
Odds ratio, > 1 km against ≤ 500 m, given terrain	GFM	1.53	1.33–1.77	182 AOIs
Odds ratio, > 1 km against ≤ 500 m, given terrain	EMS	1.32	1.12–1.56	182 AOIs
Odds ratio, water-related names, ≤ 500 m	GFM	0.55	0.37–0.82	—
Odds ratio, water-related names, ≤ 500 m, unadjusted	GFM	0.52	0.33–0.82	—
Odds ratio, water-related names, all cells	GFM	0.83	0.58–1.19	—
Odds ratio, water-related names, ≤ 500 m	EMS	0.79	0.58–1.07	—
Odds ratio, water-related names, ≤ 500 m, unadjusted	EMS	0.79	0.59–1.05	—
Odds ratio, water-related names, all cells	EMS	0.81	0.61–1.09	—
water names, ≤ 500 m	GFM	0.55	0.31–0.97	—
wetland names, ≤ 500 m	GFM	0.22	0.02–2.05	—
riverine names, ≤ 500 m	GFM	0.35	0.09–1.37	—
crossing names, ≤ 500 m	GFM	0.63	0.23–1.71	—
lowland names, ≤ 500 m	GFM	1.04	0.40–2.70	—
highland names, ≤ 500 m	GFM	1.19	0.62–2.27	—
Odds ratio, water-related, ≤ 500 m, cells ≥ 90 % assessable[a]	GFM	0.60	0.42–0.86	140,051 cells
Odds ratio, water-related, ≤ 500 m, cells ≥ 90 % assessable[a]	EMS	0.79	0.51–1.24	141,056 cells
Odds ratio, water-related, ≤ 500 m, cells fully assessable[a]	GFM	0.67	0.47–0.95	91,197 cells
Odds ratio, water-related, ≤ 500 m, cells fully assessable[a]	EMS	0.67	0.40–1.11	89,616 cells
Odds ratio, water-related, ≤ 500 m, adjusted for the baseline model refitted on GFM	GFM	0.55	0.34–0.87	—
Odds ratio, water-related, ≤ 500 m, adjusted for the baseline model refitted on GFM	EMS	0.80	0.61–1.07	—
Odds ratio, water-related, ≤ 500 m, EMS image not Sentinel-1	GFM	0.48	0.24–0.98	68 AOIs
Odds ratio, water-related, ≤ 500 m, EMS image not Sentinel-1	EMS	0.77	0.46–1.28	68 AOIs
R, held-out predictions rescored	GFM	−0.003	[−0.020, +0.011][b]	equivalent-null
R, same cells	EMS	−0.005	[−0.020, +0.008][b]	equivalent-null
R, rescored, corrected Norwegian reading	GFM	−0.007	[−0.027, +0.008][b]	equivalent-null
R, same cells, corrected Norwegian reading	EMS	−0.006	[−0.025, +0.009][b]	equivalent-null
Baseline gain ΔBS(M0→M1), every model refitted on GFM labels	GFM	-6.6×10^{-4}	-1.2×10^{-2} to 8.7×10^{-3}	R undefined
Names increment ΔBS(M1c→M3n), refitted	GFM	5.7×10^{-4}	-6.6×10^{-5} to 1.7×10^{-3}	—

Table A37: Diagnostics of the GFM labels and of the near-settlement odds ratios (amendment 6). CSI: critical success index for flooded cells. Rounded coordinates: settlements whose GeoNames coordinates are whole arc-minutes.

Quantity	Value
CSI / P(GFM=1|EMS=0) / P(EMS=1|GFM=0), all	0.51 / 1.9 % / 5.6 %
CSI / P(GFM=1|EMS=0) / P(EMS=1|GFM=0), EMS image includes Sentinel-1	0.56 / 2.1 % / 3.4 %
CSI / P(GFM=1|EMS=0) / P(EMS=1|GFM=0), other radar	0.57 / 1.7 % / 2.6 %
CSI / P(GFM=1|EMS=0) / P(EMS=1|GFM=0), optical	0.41 / 1.2 % / 17.4 %
Flooded cells within 500 m of water-related names, GFM labels	43 of 1,721 cells; 20 AOIs, 16 location groups
Flooded cells within 500 m of water-related names, EMS labels, same cells	90 of 1,721 cells; 26 AOIs, 21 location groups
Flooded cells within 500 m of water-related names, EMS labels, all cells of the GFM AOIs	101 of 3,197 cells; 29 AOIs, 23 location groups
Flooded cells within 500 m of water-related names, EMS labels, all AOIs	361 of 10,082 cells; 72 AOIs, 52 location groups
Not assessable on GFM, all cells: share; mean exclusion / permanent / seasonal / no image	57 %; 40 % / 2.5 % / 0.3 % / 48 %
Not assessable on GFM, within 500 m of water-related names: share; mean exclusion / permanent / seasonal / no image	46 %; 46 % / 1.6 % / 0.1 % / 38 %
Not assessable on GFM, within 500 m of other names: share; mean exclusion / permanent / seasonal / no image	49 %; 46 % / 1.9 % / 0.2 % / 39 %
Cells mostly excluded vs assessable: HAND $\geq$10 m; built-up $\geq$0.5; trees $\geq$0.5; EMS flooded	31 % vs 0.4 %; 7.1 % vs 2.3 %; 33 % vs 6.6 %; 2.0 % vs 12 %
Odds ratio, water-related names, all cells, V1 labels	0.83 (0.59–1.17)
Odds ratio, water-related names, all cells, EMS, V1 cells	0.80 (0.60–1.07)
R, held-out predictions rescored on V1 labels	-0.002 (90 % -0.018 to 0.010), equivalent-null
Odds ratio, water-related, $\leq$500 m, EMS, all AOIs: all settlements / without rounded coordinates	1.01 (0.77–1.32) / 1.09 (0.80–1.47)
Odds ratio, water-related, $\leq$500 m, GFM: all settlements / without rounded coordinates	0.55 (0.37–0.82) / 0.53 (0.35–0.81)
Odds ratio, water-related, $\leq$500 m, EMS, GFM cells: all settlements / without rounded coordinates	0.79 (0.58–1.07) / 0.82 (0.58–1.14)

Table A38: Calibration of the held-out predictions (exploratory, amendment 8): intercept and slope of a logistic regression of the labels on the logit of the prediction, Brier score with the reliability and resolution terms of the decomposition of Murphy (1973) (ten equal-count bins), and median within-AOI area under the ROC curve. M1: baseline model; M1s: recalibrated; M1c: with the regional composition of names (reference of the primary comparison); M3n: with the names. Skill: each AOI predicted flooded on as many cells as the labels flood (CSI: critical success index). The primary, country, time and language-model splits use the confirmatory run (frozen code); the Brier scores of the corrected run, which differ in the fourth decimal, are in Table A14.

Split	Model	Intercept	Slope	Brier	Reliability	Resolution	AUC
Primary	M1	-0.27	0.85	0.0437	3.4×10^{-4}	7.7×10^{-3}	0.90
	M1s	0.09	1.05	0.0434	5.9×10^{-5}	7.7×10^{-3}	0.90
	M1c	-0.23	0.93	0.0431	2.1×10^{-4}	8.0×10^{-3}	0.90
	M3n	-0.23	0.93	0.0431	2.2×10^{-4}	8.0×10^{-3}	0.90
Countries held out	M1	-0.57	0.69	0.0494	1.6×10^{-3}	5.5×10^{-3}	0.90
	M1s	-0.18	0.95	0.0479	6.7×10^{-4}	5.4×10^{-3}	0.90
	M1c	-0.39	0.84	0.0485	8.1×10^{-4}	5.5×10^{-3}	0.90
	M3n	-0.39	0.84	0.0485	7.9×10^{-4}	5.5×10^{-3}	0.90
Floods from 2024	M1	-0.88	0.75	0.0472	2.9×10^{-3}	3.4×10^{-3}	0.90
	M1s	-0.58	0.95	0.0460	2.0×10^{-3}	3.4×10^{-3}	0.90
	M1c	-0.70	0.91	0.0468	2.6×10^{-3}	3.5×10^{-3}	0.90
	M3n	-0.71	0.91	0.0468	2.6×10^{-3}	3.5×10^{-3}	0.90
After the LM cut-off	M1	-0.69	0.80	0.0499	1.9×10^{-3}	4.2×10^{-3}	0.90
	M1s	-0.44	0.97	0.0494	1.5×10^{-3}	4.2×10^{-3}	0.90
	M1c	-0.53	0.97	0.0497	2.0×10^{-3}	4.3×10^{-3}	0.90
	M3n	-0.54	0.97	0.0498	2.0×10^{-3}	4.3×10^{-3}	0.91
Events held out whole, frozen code	M1	-0.56	0.70	0.0488	1.4×10^{-3}	5.3×10^{-3}	0.90
	M1s	-0.13	0.96	0.0477	3.7×10^{-4}	5.1×10^{-3}	0.90
	M1c	-0.69	0.74	0.0518	2.1×10^{-3}	5.4×10^{-3}	0.90
	M3n	-0.68	0.74	0.0516	2.1×10^{-3}	5.4×10^{-3}	0.90
Events held out whole, corrected code	M1	-0.56	0.70	0.0488	1.4×10^{-3}	5.3×10^{-3}	0.90
	M1s	-0.13	0.96	0.0477	3.7×10^{-4}	5.1×10^{-3}	0.90
	M1c	-0.70	0.74	0.0519	2.2×10^{-3}	5.4×10^{-3}	0.90
	M3n	-0.70	0.74	0.0518	2.2×10^{-3}	5.4×10^{-3}	0.90
GFM labels, refitted	M1	-1.13	0.38	0.0858	6.6×10^{-3}	3.5×10^{-3}	0.86
	M1s	-0.48	0.79	0.0778	8.2×10^{-4}	2.9×10^{-3}	0.86
	M1c	-1.99	0.16	0.1155	2.5×10^{-2}	2.2×10^{-3}	0.86
	M3n	-1.99	0.16	0.1150	2.5×10^{-2}	2.2×10^{-3}	0.85

Skill in flood-mapping terms, primary split, median over AOIs (interquartile range)	
Baseline model (M1)	CSI 0.13 (0.03–0.27); hit rate 0.23 (0.07–0.42); false alarms 0.77 (0.58–0.93); AUC 0.90 (0.80–0.96); 479 AOIs
HAND alone (mean HAND of the cell; corrected run)	CSI 0.08 (0.00–0.21); hit rate 0.16; false alarms 0.84; 479 AOIs

How the names were read

Table A39: Checks of the lexicon and of the readings (exploratory, amendment 13, items B10 to B12 and C2). Counts are of the settlements nearest to at least one evaluable cell of the uncapped table, read with the frozen lexicon. Forms are matched in the ASCII field of GeoNames; Unicode names are read with umlauts, å, ø folded to one letter, æ to ae and ß to ss. Both coders' codes are in `docs/verification/v70_two_coders.csv` of the code repository.

Quantity	Value
Analysed settlements matched by homographs and wrong-language readings	
German *-ach* not in *-bach*	57
German *-ach* in *-bach*	427
Norwegian *-ore*, *-ora*	2
of them in *Store-*, *Stora-*	0
French *mont-* as a prefix	228
of them *Montreuil*, *Montier*, *Moutier*	12
German *Ried*	75
Punjabi/Urdu *kacha*	1
Burmese *taung*	139
Belgian *-eau* read as German *-au*	9
Italian names matched by the German block	5
Iran, *ab* as a word	45
Iran, *-ab* as an ending	48
Finnish *Ala*	1
Finnish *Yli*, *Ylä*	1
Edited lexicon (amendment 13, item C2): entries removed or added (form × position)	
Two-letter affixes and English transfer endings	15
Springs and wells	145
Rapids and waterfalls	10
Homographs	15
Relative qualifiers	181
Riverine entries named as ordinary meadows, islands or sand	37
Added (named omissions)	13
Entries before / after	2,538 / 2,148
Water-related readings gained / lost when the Unicode name is read	
bg	0 / 1
de	4 / 0
fi	4 / 0
km	0 / 57
pl	0 / 2
sv	17 / 0
tr	0 / 3
Agreement of the two coders on water-related derivations (κ)	0.91
Codes deposited (names in an audit sample left out)	360 of 462
Language-model reading by lexicon block: settlements, water-related, unknown	
fr	13,745, 15 %, 2.0 %
km	10,702, 39 %, 1.1 %
my	9,547, 69 %, 1.8 %
de	9,463, 21 %, 0.6 %
en	8,963, 20 %, 2.4 %
it+de	6,552, 16 %, 1.3 %
es+nah	5,569, 13 %, 0.8 %
fa+hi	5,332, 9.9 %, 4.1 %

Table A40: The multilingual lexicon by block: languages covered, countries mapped to the block, number of distinct forms per class (water, wetland, riverine, crossing, lowland, highland), match positions used and number of spring and well forms removed in the strict reading. Countries are mapped to the blocks of their main languages; regional languages are not read.

Block	Languages	Countries	Wa	We	Ri	Cr	Lo	Hi	Positions	Spr
en	English	AU BW CA CM ET FJ GB GH GY IE IM KE LK LR LS MT MW NA NG NZ PG PH SG SL SS SZ TZ UG US ZA ZM ZW	25	16	16	9	18	25	word prefix suffix	6
de	German	AT BE CH DE IT LI LU	11	12	11	12	8	15	word prefix suffix	5
nl	Dutch	BE NL SR	9	7	11	5	4	6	word prefix suffix	0
fr	French	BE BF BI BJ CA CD CF CG CH CI CM DJ DZ FR GN HT LU MA MC MG ML NE RW SN TD TG TN	15	16	15	8	12	18	word prefix suffix	5
it	Italian	CH IT SM VA	19	16	14	6	13	16	word prefix	6
es	Spanish	AD AR BO CL CO CR CU DO EC ES GT HN MX NI PA PE PR PY SV UY VE	16	22	16	6	13	22	word	6
pt	Portuguese	AO BR CV GW MZ PT TL	18	17	12	7	8	14	word	5

Block	Languages	Countries	Wa	We	Ri	Cr	Lo	Hi	Positions	Spr
el	Greek (romanized)	CY GR	24	8	3	5	5	12	word prefix	11
sh	Serbo-Croatian / Bosnian (Latin)	BA HR ME RS XK	14	11	7	6	7	14	word	7
sl	Slovene	SI	7	6	5	2	5	12	word	3
bg	Bulgarian / Macedonian (romanized)	BG MK	9	6	3	2	7	13	word	6
sq	Albanian	AL MK XK	13	8	4	5	7	11	word	4
ro	Romanian	MD RO	17	10	10	8	9	14	word	5
hu	Hungarian	HU	8	6	4	4	4	7	word prefix suffix	2
cs	Czech / Slovak	CZ SK	15	13	8	6	8	15	word	5
pl	Polish	PL	13	15	12	5	6	14	word	4
uk	East Slavic (Ukrainian / Russian / Belarusian, romanized)	BY GE KG KZ MD RU TJ TM UA UZ	15	12	15	8	14	19	word	6
lt	Lithuanian	LT	9	6	6	4	5	11	word suffix	2
lv	Latvian	LV	7	4	6	3	4	5	word suffix	2
et	Estonian	EE	7	5	4	3	3	5	word suffix	1
fi	Finnish	FI	10	8	4	3	4	10	word suffix	1
sv	Swedish	FI SE	10	7	9	6	6	16	word suffix	2
no	Norwegian (also used for Icelandic placeholders)	IS NO	16	8	14	6	6	20	word suffix	1
da	Danish	DK	6	4	5	5	4	8	word suffix	1
tr	Turkish	CY TR	11	5	3	3	6	10	word prefix suffix	4
ar	Arabic (romanized; Levant, Gulf, Maghreb conventions)	AE DJ DZ EG ER IL IQ JO LB LY MA OM PS SA SD SO SY TD TN YE	19	12	8	6	13	16	word	8
fa	Persian / Urdu / Pashto / Sindhi (romanized)	AF IR PK TJ	26	8	11	7	14	16	word prefix suffix	4
hi	Hindi / Bengali / Nepali (romanized)	BD IN NP PK	17	8	11	4	8	15	word suffix	0
vi	Vietnamese (asciiname; diacritics stripped — only low-collision syllables)	VN	7	3	2	3	1	2	word	1
id	Indonesian / Malay	ID MY SG TL	13	9	7	7	6	8	word prefix	1
tl	Filipino / Tagalog and related	PH	7	4	5	4	4	7	word	1
sw	Swahili	BI CD KE RW TZ UG	7	5	2	4	2	6	word	2
bantu	Southern/Central Bantu (Chichewa/Nyanja, Shona, Sena, Zulu/Xhosa, Sotho; broad set)	AO BW CD LS MW MZ SZ ZA ZM ZW	15	4	2	4	3	10	word prefix	1
af	Afrikaans	NA ZA ZW	6	5	3	3	5	7	word suffix	2
mg	Malagasy	MG	6	4	3	2	4	10	word prefix	1
qu	Quechua / Aymara (Andes)	BO EC PE	11	2	0	0	2	9	word suffix	2
nah	Nahuatl (Mexico)	MX	3	0	0	0	0	3	word suffix	1
my	Burmese (romanized)	MM	4	0	1	0	0	3	word	0
th	Thai / Lao (romanized)	LA TH	7	4	2	2	3	7	word	0
km	Khmer (romanized)	KH	9	0	4	3	0	4	word	0
zh	Chinese (pinyin; only low-ambiguity syllables)	CN TW	2	0	0	1	0	1	word suffix	0
ka	Georgian (romanized)	GE	5	1	0	1	3	4	word suffix	2
kk	Kazakh / Kyrgyz / Uzbek / Turkmen / Mongolian (romanized; Turkic core)	KG KZ MN TM UZ	12	5	5	5	5	9	word suffix	2
so	Somali	DJ SO	7	3	1	2	3	4	word	1
am	Amharic / Tigrinya (romanized)	ER ET	7	2	0	2	4	5	word	2
ng	Hausa / Yoruba / Igbo (Nigeria, Niger)	NE NG	8	3	0	2	1	6	word prefix	0
tupi	Tupi-derived Brazilian toponymy (prefixes only, high-precision)	BR	7	0	0	0	0	3	word prefix	0

Table A41: The 20 water-related forms that matched the nearest name of the most analysed cells (amendment 6): language block, form, position, class, number of cells and of settlements. Overlapping forms are each counted: every name in *-bach* also matches *-ach*.

Block	Form	Position	Class	Cells	Settlements
en	*creek*	word	water	4,979	62
fa	*chah*	word	water	4,665	110
en	*bridge*	word	crossing	4,605	142
en	*lake*	word	water	3,002	35
de	*ach*	suffix	water	2,891	404
de	*au*	suffix	riverine	2,701	182
de	*bach*	suffix	water	2,541	360
en	*dale*	suffix	lowland	2,515	56
hi	*char*	word	riverine	2,303	149
en	*ford*	suffix	crossing	2,161	74
en	*lower*	word	lowland	1,679	39
de	*nieder*	prefix	lowland	1,260	99
en	*vale*	word	lowland	1,234	15
ro	*lunca*	word	riverine	1,211	7
en	*flat*	word	riverine	1,092	7
sv	*dalen*	suffix	lowland	1,067	11
en	*burn*	suffix	water	1,038	30
en	*well*	suffix	water	996	47
pt	*vale*	word	lowland	881	95
no	*foss*	suffix	water	820	8

Table A42: Audit of matched forms (exploratory, amendment 8): forms that the lexicon's own rules excluded (two-letter affixes), English suffixes that also read transferred and personal names, and regional words, with the number of analysed settlements (94,524 in all) whose name they match, the cells nearest those settlements, the countries and examples (most cells first).

Block	Form	Settlements	Cells	Countries	Examples
de	*au* (suffix)	182	2,701	DE 162, AT 12, BE 8	Choerau, Sandau, Obslau, Trebbichau
sv	*as* (suffix)	158	3,282	SE 156, FI 2	Fryksas, Neistenkangas, Svartnas, Marnas
no	*as* (suffix)	3	127	NO 3	Kjonnas, Raipas, Gevingas
fa	*ab* (suffix)	73	589	PK 40, IR 33	Arab, Goth Qutab Brahui, Haji Tayyab, Haji Gulab
en	*burn* (suffix)	30	1,038	GB 22, US 6, AU 2	Heyburn, Woodburn, Wythburn, Blackburn
en	*bourne* (suffix)	5	99	GB 4, IE 1	Cranbourne, Pangbourne, Barbourne, Nailsbourne
en	*borne* (suffix)	1	30	AU 1	Tichborne
en	*pool* (suffix)	5	107	GB 4, US 1	Glenpool, Sampool, Bathpool, Firepool
en	*well* (suffix)	47	996	GB 26, US 14, IE 6, AU 1	Brauncewell, Howell, Greenwell Point, Cranwell
en	*wells* (suffix)	1	138	GB 1	Kingswells
hi	*char* (word)	149	2,303	BD 149	Char Kodalia, Char Gobindapur, Char Gazi Para, Sonatani Char
fa	*bela* (word)	9	200	PK 9	Bela Ali Khanana, Kot Bela, Bela Bhagwanpura, Makki Bela
fa	*kacha* (word)	1	11	PK 1	Kacha Pakka Mor
fa	*pattan* (word)	3	133	PK 3	Pattan Khurd, Bahkan Pattan, Sahuka Pattan
fa	*pattan* (prefix)	1	10	PK 1	Chah Pattanwala
fa	*tibba* (word)	9	79	PK 9	Tibba, Tibba Ghhale, Tibba Hussain Shah, Gharianwala Tibba

Most frequent matched forms by country (class; settlements / cells)

AU	*creek* (water; 33/4,374); *lake* (water; 9/2,573); *hill* (high; 30/2,442); *bridge* (crossing; 8/1,468); *dale* (low; 14/1,377)
DE	*ach* (water; 392/2,809); *berg* (high; 417/2,715); *au* (riverine; 162/2,642); *bach* (water; 354/2,530); *ober* (high; 213/1,419)
GB	*hill* (high; 142/3,049); *bridge* (crossing; 47/1,900); *ford* (crossing; 50/1,219); *dale* (low; 28/997); *ridge* (high; 25/860)
SE	*as* (high; 158/3,186); *berg* (high; 70/2,198); *asen* (high; 19/1,876); *berget* (high; 16/1,409); *dalen* (low; 14/1,080)
US	*hill* (high; 45/694); *creek* (water; 28/604); *valley* (low; 22/565); *ford* (crossing; 29/519); *ridge* (high; 31/443)
IT	*monte* (high; 65/1,277); *ponte* (crossing; 80/755); *piana* (low; 3/629); *poggio* (high; 21/561); *sotto* (low; 19/423)
FR	*mont* (high; 206/1,710); *haut* (high; 30/379); *pont* (crossing; 61/331); *haute* (high; 14/295); *motte* (high; 12/278)
PT	*monte* (high; 46/1,648); *vale* (low; 93/849); *cima* (high; 57/552); *agua* (water; 10/388); *porto* (crossing; 31/356)
NO	*foss* (water; 8/820); *berget* (high; 4/804); *berg* (high; 10/698); *dal* (low; 6/542); *botn* (low; 2/335)
RO	*lunca* (riverine; 7/1,211); *sus* (high; 9/818); *jos* (low; 10/744); *deal* (high; 7/715); *valea* (low; 9/503)
IR	*chah* (water; 35/3,421); *bala* (high; 15/396); *nahr* (water; 17/243); *koh* (high; 10/234); *ab* (water; 62/209)
BO	*loma* (high; 5/821); *pampa* (low; 11/531); *puentes* (crossing; 1/528); *pata* (high; 8/414); *baja* (low; 14/397)
ES	*arriba* (high; 2/285); *rio* (water; 12/284); *arroyo* (water; 3/255); *sierra* (high; 5/246); *cuesta* (high; 4/205)
MG	*ambohi* (high; 40/1,879); *ambato* (high; 14/628); *andrano* (water; 20/407); *vohi* (high; 4/261); *berge* (riverine; 1/7)
PE	*cerro* (high; 2/934); *alto* (high; 7/852); *cocha* (water; 3/336); *pampa* (low; 2/241); *loma* (high; 4/229)

Table A43: Examples of names read as water-related by the lexicon, with the derivation recorded on Wikipedia (amendment 6): for each class, the two lowest-numbered coded names whose recorded derivation matches the class and the lowest-numbered one whose derivation does not. Quotations abridged; sources in the code repository.

Class	Agreement	Name	Recorded derivation	Quotation
water	agrees	Windermere (GB)	water, other	"Windermere" is thought to translate as "Winand or Vinand's lake" ... 'mere', meaning 'lake' or 'pool'
water	agrees	Odolena Voda (CZ)	other, water	The name means "Odolen's water" ... named after Knight Odolen ... refers to the sufficient water in the area
water	disagrees	New Wells (US)	other	Austrian settlers referred to the community as Neu Wels, after the city in Austria, which was later Americanized to New Wells
wetland	agrees	Spoerkelnbruch (DE)	other, wet	Vermutlicher Namensgeber ist der Faulbaum ... Bruch ist eine alte Bezeichnung für ein Moor oder Feuchtgebiet
wetland	agrees	Clairmarais (FR)	wet	l'oïl clair + maresc : le « marais aux eaux claires »
wetland	disagrees	Spodnje Blato (SI)	low	Spodnje Blato (literally 'lower Blato')
riverine	agrees	L'Ile-Saint-Denis (FR)	riverine, other	L'Île-Saint-Denis doit son nom à l'abbaye de Saint-Denis ... qui eut seigneurie sur l'île centrale
riverine	agrees	Ostrvo (RS)	riverine	"Ostrovo" means "island" in Serbian. This name originates from the fact that Ostrovo is located on former Danube island
riverine	disagrees	Veliki Otok (SI)	water	Voda (reka Pivka) pa naj bi odtekala v Postojnsko jamo ... odtok se imenuje Veliki odtok ... dali vasi ime Veliki Otok
crossing	agrees	Pont-Pean (FR)	crossing, other	le toponyme Pont-Péan peut donc signifier « pont paysan » ou « pont construit ou possédé par un dénommé Péan »
crossing	agrees	Saint-Fargeau-Ponthierry (FR)	crossing, other	Saint-Fargeau est une déformation du nom de saint Ferréol ... Ponthierry trouverait son origine dans un pont sur l'École franchi par Thierry
crossing	disagrees	Ennerdale Bridge (GB)	low, other, water	The name Ennerdale means 'Anund's valley' ... cross- influenced by 'Ehen', the name of the local river
lowland	agrees	Dol pri Ljubljani (SI)	low	The name is derived from the Slovene common noun dol 'small valley'
lowland	agrees	Spodnji Velovlek (SI)	low	The name Spodnji Velovlek literally means 'lower Velovlek'
lowland	disagrees	Germigny-sous-Coulombs (FR)	other	Du latin germinare, « germer », du latin germinius, « rejet (botanique) » ou « rejeton, progéniture ». Coulombs fait référence à Coulombs-en-Valois

Table A44: The readings in more detail (amendment 6). Precision against derivations recorded on Wikipedia by the position of the matched water-related form and by class (a name can match at more than one position, so the position rows overlap and their counts sum to more than the names); recall by class is unweighted and therefore not comparable with the weighted recall reported in the paper. Lower part: readings by continent and languages that are not read.

Quantity	Count	Share
Precision, matched as a whole word	51/71	72 %
Precision, matched as an ending	25/30	83 %
Precision, matched as a beginning	7/8	88 %
Class water: precision / recall (unweighted)	31/38 / 31/57	82 % / 54 %
Class wetland: precision / recall (unweighted)	3/6 / 3/15	50 % / 20 %
Class riverine: precision / recall (unweighted)	8/11 / 8/13	73 % / 62 %
Class crossing: precision / recall (unweighted)	16/18 / 16/21	89 % / 76 %
Class lowland: precision / recall (unweighted)	18/37 / 18/34	49 % / 53 %
Class highland: precision / recall (unweighted)	8/15 / 8/40	53 % / 20 %
Strict reading: precision / recall (weighted)	—	83 % / 19 %
Water codes reclassified as springs or wells only	9	—
Cells whose nearest name is water-related, Africa: language model / lexicon	144,265 cells	44 % / 0.5 %
Cells whose nearest name is water-related, Asia: language model / lexicon	196,090 cells	31 % / 6.7 %
Cells whose nearest name is water-related, Europe: language model / lexicon	824,854 cells	37 % / 6.3 %
Cells whose nearest name is water-related, North America: language model / lexicon	85,031 cells	17 % / 6.8 %
Cells whose nearest name is water-related, Oceania: language model / lexicon	150,806 cells	53 % / 11 %
Cells whose nearest name is water-related, South America: language model / lexicon	106,440 cells	30 % / 3.1 %
Cells nearest a settlement in ES-Catalonia	10,000	0.7 %
Cells nearest a settlement in ES-Basque Country	174	0.0 %
Cells nearest a settlement in ES-Galicia	1,528	0.1 %
Cells nearest a settlement in ES-Valencia	20,346	1.3 %
Cells nearest a settlement in ES-Navarre	1,113	0.1 %
Cells nearest a settlement in FR-Brittany	8,229	0.5 %
Cells nearest a settlement in FR-Corsica	9,905	0.7 %
Cells nearest a settlement in IE	19,576	1.3 %
Cells nearest a settlement in AU	145,844	9.7 %
Cells whose nearest name has a riverside construction	—	0.3 %

Table A45: Recorded etymologies on Wikipedia by lexicon class and by language of the name (post hoc, amendment 5). Upper part: among the names in each class of the lexicon whose Wikipedia section states a derivation, the share whose recorded derivation is in the same class and the share whose derivation is water-related in any class (the highland class is not water-related). Lower part, for languages with at least 30 coded sections: the share of water-related derivations among the names that the lexicon read as water-related (its precision) and among the other names, with counts. Shares are unweighted; intervals are 95 % (Wilson, 1927).

Lexicon class	Coded	Same class	Water-related, any class
water	38	82 % (67–91)	84 % (70–93)
wetland	6	50 % (19–81)	100 % (61–100)
riverine	11	73 % (43–90)	82 % (52–95)
crossing	18	89 % (67–97)	94 % (74–99)
lowland	37	49 % (33–64)	54 % (38–69)
highland	15	53 % (30–75)	27 % (11–52)
Language of the name	**Coded**	**Water-related, read as water-related**	**Water-related, others**
English	48	75 % (55–88); 18/24	29 % (15–49); 7/24
French	120	67 % (50–80); 22/33	18 % (12–28); 16/87
Polish	35	40 % (12–77); 2/5	3 % (1–17); 1/30
Slovene	54	94 % (73–99); 16/17	16 % (8–31); 6/37

Table A46: A second coder of the Wikipedia excerpts (exploratory, amendment 8): a language model of another developer that took no part in the lexicon or the first coding, given the same guide and blind to the readings. Precision and recall weight the two sampling strata by their size. The rule of amendment 8: if either rate differed by 10 percentage points or more between the coders, the paper would say that the measured rates depend on the coder; they did not. Agreement between two models is not a validation by toponymists.

Quantity	Value
Excerpts coded by both; stating a derivation, agreement and κ	462; 92 %, 0.76
Water-related derivation, agreement and κ (350 excerpts)	96 %, 0.91
κ by class	water 0.94, wet 0.97, riverine 0.79, crossing 0.97, low 0.89, high 0.90, other 0.69
Lexicon, broad: precision; recall (coder 1 on the same excerpts; coder 2)	76 %; 22 % — 74 %; 18 %
Language model, broad: precision; recall (coder 1; coder 2)	41 %; 61 % — 46 %; 60 %
Lexicon, strict: precision; recall (coder 2)	76 %; 15 %
Names read as water-related with a water-related derivation in coder 2's codes: transferred or commemorative; feature not at the site	10 of 84; 3 of 84

Table A47: Classes that the language-model reading (N-read) assigned to the nearest names of the analysed cells (exploratory; amendment 10, item B15), counted once per name and weighted by the number of analysed cells nearest each name. "Unknown": the model gave no class.

Class	Names	Share	Cells	Share
water	12,072	13 %	234,593	16 %
wetland	7,602	8.0 %	125,945	8.4 %
riverine	7,716	8.2 %	105,543	7.0 %
crossing	1,783	1.9 %	29,676	2.0 %
lowland	3,472	3.7 %	61,728	4.1 %
highland	8,832	9.3 %	161,156	11 %
other environmental	10,937	12 %	148,830	9.9 %
non-environmental	38,219	40 %	550,235	37 %
unknown	3,891	4.1 %	89,780	6.0 %

Table A48: The nearest settlements of the analysed cells by GeoNames feature code (exploratory; amendment 11, item B9): settlements, their share, those read as water-related and the cells nearest them. PPL: populated place; PPLA–PPLA4: seats of administrative divisions; PPLX: section of a populated place; PPLH: historical; PPLQ: abandoned; PPLW: destroyed; PPLL: locality; PPLF: farm village; PPLR: religious.

Feature code	Settlements	Share	Water-related	Cells
PPL	84,867	90 %	5369	1,290,881
PPLX	3,907	4.1 %	278	70,588
PPLL	1,807	1.9 %	178	37,817
PPLA3	1,669	1.8 %	59	54,253
PPLQ	633	0.7 %	33	7,390
PPLA2	598	0.6 %	26	25,149
PPLA4	579	0.6 %	84	12,093
PPLF	165	0.2 %	16	1,726
PPLA	159	0.2 %	9	5,234
PPLH	82	0.1 %	3	1,780
PPLCH	20	0.0 %	0	151
PPLW	18	0.0 %	1	138
PPLR	11	0.0 %	0	172
PPLC	4	0.0 %	0	102
PPLS	4	0.0 %	0	10
PPLG	1	0.0 %	0	2
Settlements whose coordinates are whole arc-minutes: 14 %				

Table A49: Settlements whose namesake is recorded on Wikidata (property P138), classed by the type of the namesake (amendment 5). The lexicon contains no river names by design, so it reads few settlements named after a river; most recorded namesakes are in France.

Quantity	Count
Analysed settlements	94,524
with a Wikidata item	26,509
with a recorded namesake (P138)	299
named after a water feature	139
named after a person, event or other commemoration	97
Water-feature namesakes read as water-related by the lexicon	9 of 139 (6 %)
Same, by the language model	66 of 139 (47 %)
Lexicon water-related names with a water-feature namesake	9 of 13 (69 %)
Language-model water-related names with a water-feature namesake	66 of 92 (72 %)
Lexicon water-related names with a commemorative namesake	2 of 13 (15 %)

Table A50: Precision and recall of the lexicon's water-related reading by language of the article, for languages with at least 30 coded names (amendment 5; recall by language, amendment 7, addendum 1). Recall is estimated as a domain with the design weights of the whole sample (stratum size over sample size), whereas Table A51 weights the strata by their sizes within each country, with percentile intervals from 2,000 resamples within strata. The derivations were coded by the language model that compiled the lexicon. Strict: the strict reading (water without springs and wells, wetland, riverine) against strict derivations (amendment 8, addendum 7).

Language	Coded names	Precision (95 % interval)	Strict precision	Recall (95 % interval)
fr	120	22 of 33 (67 %; 50 %–80 %)	5 of 7 (36 %–92 %)	17 % (10 %–28 %)
sl	54	16 of 17 (94 %; 73 %–99 %)	8 of 9 (56 %–98 %)	29 % (15 %–59 %)
en	48	18 of 24 (75 %; 55 %–88 %)	9 of 11 (52 %–95 %)	28 % (16 %–56 %)
pl	35	2 of 5 (40 %; 12 %–77 %)	1 of 1 (21 %–100 %)	23 % (0 %–100 %)

Table A51: Precision and recall of the lexicon's water-related reading by country, against etymologies recorded on Wikipedia (exploratory, amendment 10, item B6): countries with at least ten coded names. Recall weights the two sampling strata by their sizes within the country, whereas Table A50 uses the design weights of the whole sample (stratum size over sample size); the two agree on precision and differ on recall where the sampling fraction differs between countries. Forms: the only matched forms with at least ten coded names read as water-related. Both coders are language models; coder 1 compiled the lexicon.

Country	Coded	Read as water-related	Precision (1)	Recall (1)	Precision (2)	Recall (2)
FR	120	33	67 %	12 %	61 %	7.9 %
SI	54	17	94 %	21 %	85 %	19 %
PL	35	5	40 %	17 %	80 %	12 %
GB	33	16	81 %	20 %	83 %	23 %
CZ	20	6	83 %	33 %	83 %	20 %
MX	10	8	88 %	100 %	88 %	100 %
Form *sous* (coder 1)		12	25 %			
Form *sous* (coder 2)		11	27 %			

Table A52: Analysed cells whose names are read in a language other than their own (exploratory, amendment 10, item B14). Country level: the first language of the GeoNames country record is not among the lexicon blocks applied to the country. Regions: Natural Earth first-level units (point in polygon of the cell centre).

Country or region	Cells	Water-related	Flooded, 500 m	AOIs
Countries whose first language is not among the lexicon blocks applied				
PK (ur; read as fa, hi)	61,219	2,249	4534	
BD (bn; read as hi)	19,720	2,652	643	
LK (si; read as en)	9,032	0	147	
MK (mk; read as bg, sq)	5,132	256	0	
TL (tet; read as id, pt)	4,962	0	33	
RU (ru; read as uk)	2,070	0	0	
SK (sk; read as cs)	1,926	52	0	
LA (lo; read as th)	333	9	0	
IN (en; read as hi)	323	0	0	
LU (lb; read as de, fr)	6	0	0	
Regions with a language other than the one read				
Wales	0	0	0	0
Highland and Eilean Siar	3,997	348	0	2
Northern Ireland	3,316	0	0	2
Ireland	18,290	2,220	21	19
Catalan-speaking Spain	30,255	93	58	5
Basque Country and Navarre	1,230	3	1	4
Galicia	1,528	0	0	1
Brittany	7,943	244	6	7
Alsace	0	0	0	0
Friesland	0	0	0	0
Finnmark and Troms	3,818	0	2	2
Norrbotten	2,621	0	11	4
Lapland (Finland)	11,124	1,124	11	5

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